

Title: A Randomized, Double Blind Placebo Controlled Study to Evaluate the Safety, Tolerability, and Pharmacokinetics of Multiple Ascending Subcutaneous Doses of AMG 570 in Subjects With Rheumatoid Arthritis

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Name of Investigator

Date (DD Month YYYY)

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Protocol Synopsis

Title: A Randomized, Double Blind Placebo Controlled Study to Evaluate the Safety, Tolerability, and Pharmacokinetics of Multiple Ascending Subcutaneous Doses of AMG 570 in Subjects with Rheumatoid Arthritis

Study Phase: 1b

Indication: Rheumatoid Arthritis

Primary Objective:

- To evaluate the safety and tolerability of AMG 570 following multiple dose subcutaneous (SC) administration in subjects with rheumatoid arthritis (RA).

Secondary Objective(s):

- To characterize the pharmacokinetic profile of AMG 570 following multiple dose SC administration in subjects with RA.

Exploratory Objective(s):

- To characterize the pharmacodynamic effects of multiple subcutaneous doses of AMG 570 in subjects with RA on peripheral blood B cell levels (total B cells as well as naive and memory subpopulations), peripheral blood ICOSL Receptor Occupancy on B cells, serum BAFF levels, serum Immunoglobulin levels, plasma cells and circulating T follicular helper (Tfh) cells.
- To characterize pharmacokinetic and pharmacodynamic relationships after multiple subcutaneous doses of AMG 570 in subjects with RA.
- To characterize effects of multiple subcutaneous doses of AMG 570 on clinical and serologic indices of RA disease activity in subjects with RA.
- To evaluate the immunogenicity of multiple subcutaneous doses of AMG 570 in subjects RA.

Hypotheses:

Multiple SC dose administration of AMG 570 will have acceptable safety and tolerability profiles in RA subjects within the proposed dose range (70 to 420 mg AMG 570)

Primary Endpoint:

Subject incidence of treatment-emergent adverse events (including clinically significant changes in physical examinations), vital signs, laboratory safety tests, and electrocardiograms (ECGs).

Secondary Endpoint(s):

- AMG 570 Serum PK profiles after multiple dose administrations. PK parameters will include, but are not limited to, maximum observed concentration [C_{max}] and area under the concentration-time curve [AUC] after the first and the last doses, and the accumulation ratio.

Exploratory Endpoint(s):

- Peripheral blood ICOSL receptor occupancy.
- Baseline levels, post-dose levels and change from baseline in peripheral blood percentage and absolute counts of total T cells, B cells and NK cells.
- Baseline levels, post-dose levels and change from baseline in peripheral blood percentage and absolute counts of naive and memory B cells (naïve = CD19+IgD+CD27-, memory = CD19+IgD-CD27+)
- Baseline levels, post-dose levels and change from baseline in circulating Plasma cells by gene signature

- Baseline levels, post-dose levels and change from baseline in circulating Tfh by flow cytometry and gene signature
- Baseline levels, post-dose levels and change from baseline in serum Immunoglobulin M, Immunoglobulin G and Immunoglobulin A
- Baseline levels, post-dose levels and change from baseline in serum BAFF
- Baseline levels, post-dose levels and change from baseline in serum CRP, ESR, IgG+ Rheumatoid Factor, IgM+ Rheumatoid Factor and anti-citrullinated protein antibody
- Changes from baseline over time in the disease activity score based on 28 joints (DAS 28-CRP)
- Changes from baseline over time in Patient Global Assessment of Disease activity
- Changes from baseline over time in Physician Global Assessment of Disease activity
- Anti-AMG 570 binding antibodies

Study Design: This is a randomized, placebo-controlled, double-blind, multiple ascending dose (MAD) study in RA subjects. The study consists of 4 SC cohorts with SC doses ranging from 70 to 420 mg. Subjects will be randomized in a 3:1 ratio to receive AMG 570 or placebo accordingly.

Escalation to a higher dose cohort will be contingent on a review indicating that the previous dose regimen(s) has (have) been found to demonstrate an acceptable safety and tolerability profile at a dose level review meeting (DLRM). The DLRM will review blinded safety and laboratory data once at least 6 subjects in preceding cohorts have reached day 29. Safety data from the ongoing single ascending dose (SAD) study with AMG 570 (20140322) will also be incorporated to ensure that any higher dose cohort in the MAD study has already been found to have an acceptable safety and tolerability profile in the corresponding dose cohort of the SAD study.

The 140 mg Q2W, 210 mg Q2W, or 420 mg Q2W doses may be revised based on the data received from the earlier cohorts in this study, the higher doses evaluated in the ongoing FIH SAD study in healthy subjects, as well as revised PK/PD predictions (if any) based on emerging data from the SAD or MAD study.

Sample Size: Approximately 32 subjects will be enrolled in 4 cohorts with 6 active and 2 placebo in each cohort.

Summary of Subject Eligibility Criteria: Subjects in this study will be males and females with RA between ages of 18-75 years (inclusive). Subjects should have a body mass index of ≥ 18 and ≤ 35 kg/m². Subjects should be up to date on immunizations with status determined by the site investigator. For a full list of eligibility criteria, please refer to [Section 4.1](#) and [Section 4.2](#).

Dose Levels

Cohort #	Planned Dose	Route	N (active:placebo)
1	70 mg q 2 wk x 6 doses	SC	8 (6:2)
2	140 mg q 2 wk x 6 doses	SC	8 (6:2)
3	210 mg q 2 wk x 6 doses	SC	8 (6:2)
4	420 mg q 2 wk x 6 doses	SC	8 (6:2)

Investigational Product

Amgen Investigational Product Dosage and Administration: AMG 570 will be packaged in 5-ml clear glass vials containing 1 ml of 70 mg/ml AMG 570 formulated as below. Placebo will be presented in similar containers and stored the same as AMG 570. Vials will be stored between 2°C to 8°C.

For Cohorts 1 to 3, doses will be administered in a 1.0 mL per injection format (up to 3 x 1 mL injections depending on treatment arm). For Cohort 4, doses will be administered as a 1.5 mL per injection format (4 x 1.5 mL).

Dosage and IP Administration

Product Name	AMG 570	Placebo
Formulation Description	10 mM acetate, 9.0% (w/v) sucrose, 0.01% (w/v) polysorbate 80	10 mM acetate, 9.0% (w/v) sucrose, 0.004% (w/v) polysorbate 20
Dosage Form	Solution	Solution
Unit dose Strength	70 mg/ml	NA

Procedures:

The planned length of participation in the study will be approximately 10 months (22-26 visits) which includes up to 6 weeks for screening, 3-8 days for inpatient visit, outpatient visits up to 10 weeks for drug dosing, 24 weeks for evaluation of drug distribution after last dose, followed by end of study (EOS) visit.

Screening: After signing the informed consent, subjects will be screened to determine eligibility within a period of 42 days before dosing on day 1. Subjects who require additional vaccination will receive vaccines during the screening period and at least 28 days prior to dosing. Adverse events will be collected for a subject from the time of signed informed consent until the subject concludes participation in the study including any extended follow-up for anti-AMG 570 antibodies, B cell counts, or serum IgG levels. Concomitant medications will be collected throughout the study.

Treatment: Inclusion in the study requires pre-existing background therapy with methotrexate (or other non-biologic/non-synthetic DMARD) for ≥ 12 weeks prior to receipt of investigational product (IP), and subjects will remain on this background therapy through the length of the study.

Subjects will be admitted to the research facility on Day -1 to collect baseline clinical and laboratory data. Subjects will receive the first dose of IP on Day 1. Subjects will check out of the research facility no less than on Day 3 and up to Day 11 after all assessments have been completed. Subjects must return to the research facility on an outpatient basis at specified time points for collection of blood samples for PK and PD measurements, RA disease assessments, and completion of safety assessments.

Subjects will receive a SC dose of IP on days 1, 15, 29, 43, 57, and 71. Subjects will be followed for 24 weeks after the last dose.

End of study: A subject's participation in the study will conclude with the completion of the end of study procedures. However, if an end of study test result demonstrates a significant clinical or laboratory abnormality, the subject will be followed until resolution of the abnormality or until it is considered clinically stable by the Principal Investigator.

- If a subject's CD3-CD19+ B cell count at screening is ≥ 107 cells/ μ L at screening and is < 107 cells/ μ L at the end of study visit, the subject will return for further testing of B cell counts until values are within the healthy range. If a subject's CD3-CD19+ B cell count at screening is between 70-106 (inclusive) at screening and is less than their screening value at the end of study visit, the subject will return for further testing of B cell counts until values are greater than or equal to baseline values. If a subject's serum IgG level is < 639 mg/dl (the lower limit of normal for healthy individuals), the subject will return for further testing of serum IgG until values are within the healthy range. Such follow-up visit(s) for B cell counts and serum IgG will continue every 3 months, up to 12 months after the end of study.

For a full list of study procedures, including the timing of each procedure, please refer to [Section 7](#) and the Schedule of Assessments ([Table 4](#)).

Statistical Considerations: Descriptive statistics will be provided for selected demographics, safety, pharmacokinetics, pharmacodynamics, and biomarker data by dose, dose schedule, and time as appropriate. Descriptive statistics on continuous data will include means, medians, standard deviations and ranges, while categorical data will be summarized using frequency counts and percentages. Graphical summaries of the data may also be presented.

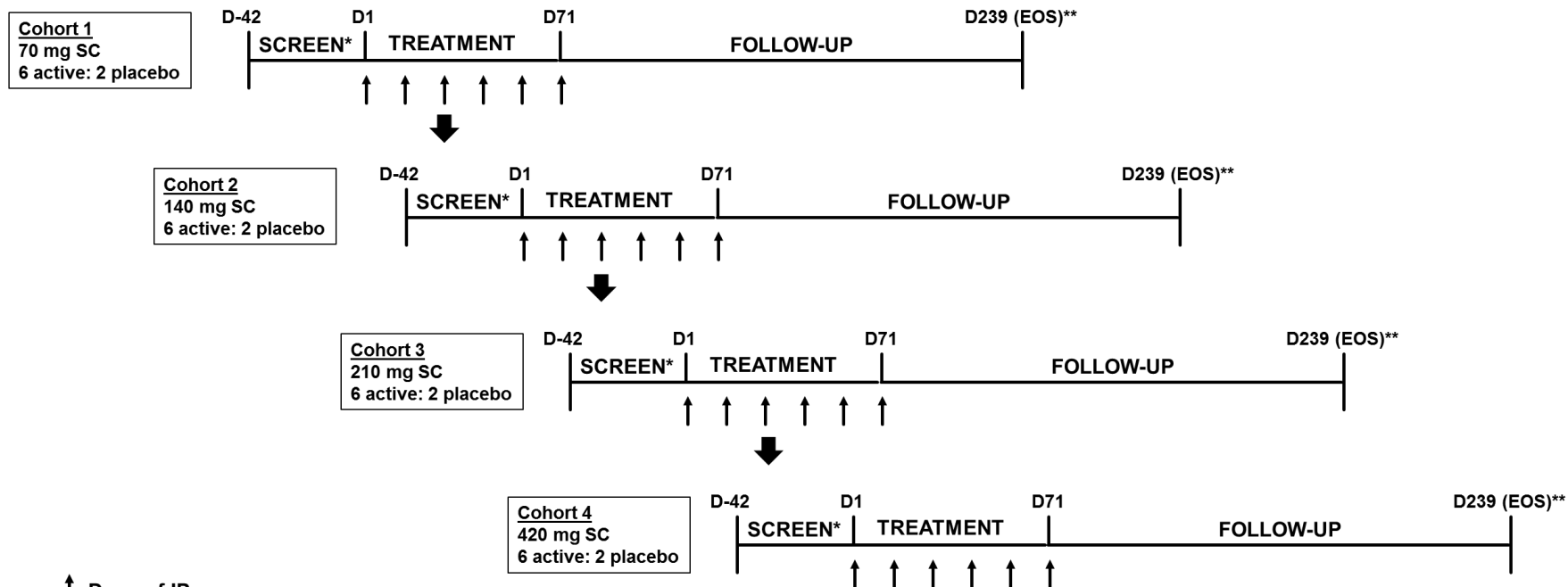
Safety data will be reviewed on an ongoing basis. Amgen, in consultation with the site investigator, will review in Dose Level Review Meetings (DLRMs) all available accumulating data, including data from the ongoing AMG 570 SAD study (20140322) prior to making dose escalation decisions.

For a full description of statistical analysis methods, please refer to [Section 10](#).

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Study Design and Treatment Schema



↑ Dose of IP

↓ DLRM (initiated once at least 6 subjects have completed the Day 29 visit)

* Patients maintained on stable MTX dose through duration of screening and treatment

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Study Glossary

Abbreviation or Term	Definition/Explanation
ACR/EULAR	American College of Rheumatology/European League Against Rheumatism
ADA	anti-drug antibodies
ALT	alanine aminotransferase
ANC	absolute neutrophil count
AST	aspartate aminotransferase
AUC	area under the concentration-time curve
BAFF	B cell activating factor
B7RP-1	B7-related protein-1 (also referred to as ICOSL)
BMI	body mass index
CI	confidence interval
C _{max}	maximum observed concentration
CRF	case report form
DCs	dendritic cells
DILI	drug-induced liver injury
DLRM	dose level review meeting
DLT	dose limiting toxicity
DMARD	disease modifying anti-rheumatic drug
ECG	electrocardiogram
eCRF	electronic case report form
EOS	end of study
EMA	European Medicines Agency
end of study for individual subject	defined as the last date that protocol-specified procedures are conducted for an individual subject
end of treatment	defined as the date of final assessment for the protocol specified treatment phase of the study for an individual subject
end of study (primary completion)	defined as the date when the last subject is assessed or receives an intervention for the purposes of final collection of data for the primary endpoint(s).
end of trial	defined as the date when the last subject is assessed or receives an intervention for evaluation in the study; if the study includes multiple parts (eg, safety follow-up), the end of study would include these additional parts
FIH	First in Human
FSH	Follicle stimulating hormone
GCP	Good Clinical Practice
GLP	Good Laboratory Practice

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Abbreviation or Term	Definition/Explanation
GSO	Global Safety Officer
HBcAb	hepatitis B core antibody
HepBsAg	hepatitis B surface antigen
HepCAb	hepatitis C antibody
HIV	human immunodeficiency virus
Ig	Immunoglobulin
IC ₅₀	inhibitory concentration at 50%
IC ₉₀	inhibitory concentration at 90%
IC ₉₉	inhibitory concentration at 99%
ICF	informed consent form
ICOS	inducible co-stimulator
ICOSL	inducible co-stimulator ligand
ICH	International Conference On Harmonisation
IV	intravenous
IP	Investigational Product
IPIM	Investigational Product Instruction Manual
IPRO	Immunophenotyping and Receptor Occupancy
IRB	Institutional Review Board
LLN	lower limit of normal
LLOQ	lower limit of quantification
MABEL	Minimum Anticipated Biological Effect Level
MAD	Multiple ascending dose
MRSD	maximum recommended starting dose
NOAEL	no observed adverse effect level
PD	pharmacodynamic
PK	pharmacokinetic
PR	the interval measured from the beginning of the P wave to the beginning of the QRS complex in the heart's electrical cycle as measured by electrocardiogram
QRS	the interval between the Q wave and the S wave in the heart's electrical cycle as measured by electrocardiogram; represents the time it takes for depolarization of the ventricles
QT	measure of the time between the start of the Q wave and the end of the T wave in the heart's electrical cycle as measured by electrocardiogram
QTc	QT interval corrected for heart rate using accepted methodology
RA	Rheumatoid arthritis
RO	receptor occupancy

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Abbreviation or Term	Definition/Explanation
RNA	ribonucleic acid
SC	Subcutaneous
SLE	Systemic Lupus Erythematosus
study day 1	defined as the first day that protocol specified investigational product is administered to the subject
Tfh	T follicular helper cell
$t_{\max}$	time to maximum concentration
ULN	upper limit of normal

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1. OBJECTIVES

1.1 Primary

To evaluate the safety and tolerability of AMG 570 following multiple dose subcutaneous (SC) administration in subjects with rheumatoid arthritis (RA).

1.2 Secondary

To characterize the pharmacokinetic profile of AMG 570 following multiple dose SC administration in subjects with RA.

1.3 Exploratory

- To characterize the pharmacodynamic effects of multiple subcutaneous doses of AMG 570 in subjects with RA on peripheral blood B cell levels (total B cells as well as naive and memory subpopulations), peripheral blood ICOSL Receptor Occupancy on B cells, serum BAFF levels, serum Immunoglobulin levels, plasma cells and circulating T follicular helper (Tfh) cells.
- To characterize pharmacokinetic and pharmacodynamic relationships after multiple subcutaneous doses of AMG 570 in subjects with RA.
- To characterize effects of multiple subcutaneous doses of AMG 570 on clinical and serologic indices of RA disease activity in subjects with RA.
- To evaluate the immunogenicity of multiple subcutaneous doses of AMG 570 in subjects RA

2. BACKGROUND AND RATIONALE

2.1 Disease

RA is a chronic inflammatory disease of the joints that can lead to significant disability due to cartilage and bone damage. RA is estimated to have a worldwide prevalence between 0.5-1%. Extra-articular manifestations involving an array of organ systems are also observed in some patients ([Smolen, 2016](#)).

RA results from the aberrant infiltration of immune cells into the joint. This infiltrate is a mix of immune cells from both the lymphoid and myeloid lineages. These cells release lipid and protein mediators, including cytokines, proteases, and antibodies, that perpetuate the inflammatory response, lead to clinical symptoms, and potentially tissue destruction ([Catrina, 2016](#)). The use and proliferation of available disease modifying anti-rheumatic drugs (DMARDs), including conventional, biologic, and targeted synthetic drugs, have allowed up to 40% of patients with moderate to severe disease achieve a low disease activity state as defined by a 70% response in the American College of Rheumatology improvement criteria (ACR70) ([Smolen, 2016](#)). Nonetheless, a number of patients remain symptomatic, and innovative therapies hold potential to further improve outcomes.

2.1.1 Autoantibodies Result From T cell-B cell Interactions

A central driver of the inflammatory response in RA is considered to be the production of autoantibodies. Autoantibody production results from maladaptive interactions between autoreactive T cells and B cells in lymphoid tissue ([Catrina, 2016](#)). In part reflecting the central role of autoantibodies in RA pathogenesis, 50-70% of RA patients have detectable autoantibodies such as rheumatoid factor (RF) or anti-citrullinated protein antibodies (ACPAs) ([Smolen, 2016](#)). ACPAs are notable as these antibodies may be directly responsible for initiating and perpetuating arthritis ([Catrina, 2016](#)). In addition, preliminary data indicate that early intervention to forestall the development of autoantibodies in individuals susceptible to RA may in turn delay the onset of disease ([Gerlag, 2016](#)). Together, these data indicate that the production of autoantibodies is a primary precipitant of the inflammatory cascade in RA and that targeting autoantibody production through disruption of T cell-B cell interactions provides a means to improve clinical outcomes.

Over the last 15 years, an increasingly clear picture of the molecular cues and temporal dynamics of T cell-B cell interactions leading to autoantibody production has emerged. In short, antigen-specific T cells survey B cell areas within lymphoid tissue, and upon encountering a B cell that displays the requisite antigen, the T cell and B cell engage in a bidirectional “crosstalk” that ultimately leads to the differentiation of the T cell into the T follicular helper (Tfh) cell lineage and activation of the B cell ([Craft, 2012](#)). Activation of autoreactive B cells initiates a differentiation program which leads to the generation of autoantibody secreting plasmablasts and plasma cells. Tfh cells are noteworthy because these cells are particularly efficient at further engaging B cells to amplify the magnitude and breadth of the antibody response. Studies in humans as well as pre-clinical animal models indicate that the T cell costimulatory molecule, inducible T cell costimulator (ICOS), and its ligand, ICOSL, (also known as B7 related protein-1 or B7RP-1), expressed on B cells and dendritic cells (DC), is central to the differentiation of T cells to the Tfh lineage as the absence of ICOS: ICOSL interactions leads to substantial loss in Tfh cells as well as antibody production ([Craft, 2012](#)).

2.1.2 Modulation of T cell-B cell Function in Rheumatoid Arthritis

Therapeutics to modulate T cell- B cell interactions and inhibit autoantibody production have focused on unilateral targeting of the T cell- B cell dyad. Abatacept inhibits T cell costimulation through blockade of the ligand for the costimulatory molecule, CD28, found on T cells. Abatacept is clinically effective in RA. B cell targeting has focused on

modulating B cell numbers to minimize the likelihood of T-B interactions. Strategies to directly deplete B cells with rituximab or neutralize B cell growth factors, such as BAFF, have been employed and studied in subjects with RA. Modulation of T cell costimulation as well as depletion of B cells have both shown clinical efficacy in RA subjects with abatacept and rituximab approved for this indication. Though rituximab and abatacept are indicated for only partially overlapping subject populations, the proportion of subjects who achieve ACR70 responses 6 months after treatment is ~10-20% (Cohen 2006, Genovese 2005, Kremer 2003). Thus, new molecular entities, such as AMG 570, that are capable of further modulating T-B interactions may provide enhanced efficacy.

2.2 Amgen Investigational Product Background

2.2.1 Pharmacology

AMG 570 is a bispecific molecule inhibiting both BAFF and ICOSL. AMG 570 contains two tandem copies of BAFF-binding peptides identical to the BAFF binding peptides of blisibimod (also known as AMG 623) fused to the C-terminus of AMG 557, a fully human IgG2 against ICOSL currently in clinical development for SLE. In terms of anti-BAFF effect, AMG 570 has similar binding affinity and cellular potency to blisibimod. AMG 570 has 29 pM K_D BAFF binding affinity and 0.86 nM IC_{50} in a BAFF cellular assay. Blisibimod was reported to have an IC_{50} of 0.2 nM. For the anti-ICOSL effect, AMG 570 has similar binding affinity and inhibitory activity to AMG 557. AMG 570 has 28 pM K_D ICOSL binding affinity and 1.36 nM IC_{50} in a ICOSL cellular assay. AMG 557 has 17 pM K_D ICOSL binding affinity and AMG 557 parental antibody 16H has 1.35 nM IC_{50} in an ICOSL cellular assay. AMG 570 binds and inhibits cynomolgus monkey BAFF and ICOSL, but not rodent ICOSL. A mouse surrogate BAFF/ICOSL bispecific molecule showed dual-target inhibition in vivo and was more efficacious than the single ICOSL or BAFF inhibitors in the mouse NZB/NZW lupus model. Thus, we hypothesize that targeting both BAFF and ICOSL by AMG 570 will achieve a large effect size in the treatment of autoantibody associated diseases, including RA and SLE.

2.2.2 Toxicology

AMG 570 has been evaluated in a series of toxicology studies in laboratory animals and in vitro test systems. The Good Laboratory Practice (GLP) 3-month repeat-dose toxicology study in the cynomolgus monkey with an 8-month recovery was performed as a pivotal toxicology study to support this clinical study. AMG 570 was administered

subcutaneously (SC) at 0, 30, 100, or 200 mg/kg once weekly for 13 weeks (4 main study animals/sex/group and 2 recovery animals/sex/group at 0 and 200 mg/kg).

Weekly SC administration of AMG 570 for 13 weeks was well tolerated up to 200 mg/kg in cynomolgus monkeys with sustained AMG 570 exposure (TK) and PD response. The no-observed-adverse-effect level (NOAEL) was 200 mg/kg, providing exposure margin of ≥ 29 (Table 1 in Section 2.4.5). Anti-drug antibodies (ADAs) to AMG 570 were detected in all dose groups, affecting AMG 570 exposure and PD response in some animals; however, sustained exposure and PD response were observed in 19 of 28 AMG 570-dosed animals (4 of 8 at 30 mg/kg, 5 of 8 at 100 mg/kg and 10 of 12 at 200 mg/kg) throughout the 13-week administration period, enabling a robust toxicology evaluation. In these animals, AMG 570 pharmacology-related effects (ie, decreased B cells, IgG and lymphoid depletion of the lymphoid tissues) and injection site reactions (attributed to a localized response to the SC administered biological protein) were observed at all dose levels. All cellular and lymphoid organ changes were reversible.

2.3 Risk Assessment

AMG 570 is currently being evaluated in an ongoing FIH single ascending dose study in healthy subjects (20140322). As of 01 June 2018, 56 subjects have received blinded investigational product at SC doses up to 700 mg. All subjects receiving dose levels up to 420 mg SC in the single ascending dose study have completed and demonstrated acceptable safety and tolerability with no severe, life-threatening, or fatal events reported. All adverse events have been of mild to moderate severity. Subjects receiving 700 mg SC remain on study with safety data and a dose level review meeting anticipated later this year. Further details may be found in Section 2.4.4.

For this multiple ascending dose study in RA patients, 17 subjects have received either 70 mg or 140 mg of blinded investigational product as of 01 June 2018. As of 01 June 2018, no deaths or drug-related serious adverse events, dose-limiting toxicities, or withdrawals due to adverse events have been reported. In general, AMG 570 was well tolerated at the doses tested.

An additional assessment of potential side effects of multiple doses of AMG 570 is based on the pre-clinical studies conducted to date and by observations of safety and tolerability of AMG 557 and blisibimod clinical studies. Summaries of findings from the pre-clinical studies with AMG 570 can be found in the [AMG 570 Investigator's Brochure \(IB\)](#). No target organs beyond pharmacologic effects of AMG 570 were identified up to

the high dose in the 3-month GLP cynomolgus monkey toxicology study (200 mg/kg: 11 to 64.5- times higher than planned clinical doses [70 to 420 mg]).

Potential side effects of AMG 570 could also be related to the biological nature of the molecule, and mode of administration.

- **Immunosuppression:** risks which may be associated with immunosuppression with reduction in humoral and cellular immunity include infection and malignancy.
- **Hypersensitivity:** all therapeutic products (small molecules and biologics) can be associated with hypersensitivity reactions. Most of the hypersensitivity reactions are mild in severity, although severe reactions may occur (eg, angioedema, anaphylaxis).
- **Injection site reactions:** (eg, erythema, itching, hematoma, swelling or bruising and pain) are common side effect of drugs with SC administration. These reactions can range from mild to severe (including injection site necrosis which is an uncommon side effect).

The proposed risk assessment and management plan for the study has been developed in accordance with the requirements of regulatory guidelines and strategies to identify and mitigate risks for clinical trials with investigational medicinal products.

Please refer to the [AMG 570 IB, Section 7](#) for a description of these potential side effects.

A newly completed 6-month toxicology study in the cynomolgus monkey has shown an increase in RBC and hemoglobin in a small number of animals at exposure levels much higher than expected in clinical subjects. The relevance of this for humans is not currently known, and hematology safety laboratory tests will be monitored for any changes.

2.4 Rationale

Given the centrality of T-B cell interactions in the production of autoantibodies, approaches to disrupt this axis hold promise in providing enhanced efficacy. AMG 570 utilizes a bispecific design to simultaneously target both members of the T cell-B cell pair. Through targeting of ICOSL, AMG 570 disrupts the productivity of T cell-B cell interactions by interfering with Tfh activity, and through targeting of BAFF, AMG 570 decreases the pool of B cells available to interact with T cells. Thus, AMG 570 seeks to impact both the qualitative and quantitative outputs of T cell-B cell collaboration to decrease autoantibody production and thereby provide enhanced efficacy compared to existing therapies.

2.4.1 AMG 570, AMG 557 and Blisibimod Molecular Structure Information

The bispecific AMG 570 antibody has a peptide-linker-peptide-linker construct fused to the C-terminus of each heavy chain of AMG 557. AMG 557 is a human IgG2. Blisibimod has a peptide-linker-peptide-linker construct fused to the N-terminus of each chain of a human Fc. The peptides in AMG 570 are BAFF-binding peptides identical to those in blisibimod. Molecular weights of AMG 570, AMG 557 and blisibimod are approximately 161, 148, and 64 kDa, respectively. Each of the constituent pathways targeted by AMG 570 have been investigated in patient populations. In total, these data indicate that in isolation, the two arms have acceptable safety and tolerability profiles and show trends indicative of clinical activity (Furie, 2014 and Sullivan, 2016). A brief description of the clinical background of each of the constituent molecules is found in Sections 2.4.2 and 2.4.3.

2.4.2 AMG 557 Clinical Background

Single dose administration of AMG 557 to subjects with mild, stable SLE with subcutaneous (SC) doses of 1.8, 6, 18, 60, 140, and IV dosing of 210 mg and 18 mg demonstrated an acceptable safety profile with no observed neutralizing antibodies (study 20060132). AMG 557 demonstrated nonlinear pharmacokinetic properties, as expected for therapeutic targeting of a cell surface receptor. The degree of target occupancy was dose/concentration-related, reversible, and achieving maximal levels in the 140 mg SC group. Multiple dose administration of AMG 557 to subjects with mild, stable SLE with SC doses of 6, 18, 30, 45, 70, 140, and 210 mg demonstrated an acceptable safety profile with no observed neutralizing antibodies (study 20060169). Consistent, dose-dependent pharmacodynamic effects on KLH responses were observed following multiple doses of AMG 557. Multiple dose administration of 210 mg SC for 6 months was also investigated in patients with active lupus arthritis (study 20101103). Though not statistically significant, trends of improvement in lupus arthritis response as well as general metrics of SLE activity (including changes in BILAG and SLEDAI) suggested clinical activity of AMG 557.

The estimated mean AMG 557 target occupancy IC_{50} was 0.0888 $\mu\text{g/mL}$ (0.60 nM), IC_{90} of 0.8 $\mu\text{g/mL}$ (5.4 nM), and IC_{99} of 8.8 $\mu\text{g/mL}$ (59.5 nM).

2.4.3 Blisibimod Clinical Background

Subjects with mild to moderate SLE were treated with blisibimod single SC doses of 0.1, 0.3, 1.0, or 3.0 mg/kg or IV doses of 1.0, 3.0, or 6.0 mg/kg in a first-in-human study (study 20040147) and multiple dose 0.3, 1.0, 3.0 (SC) or 6.0 (IV) mg/kg in phase 1b

(study 20040250). In the multiple dose study, blisibimod treatment led to a significant decrease in naïve B cells (IgD⁺CD27⁻) and an increase in class-switched memory B cells (IgD⁻CD27⁺) (Stohl et al, 2015). This significant reduction in naïve B-cells was observed when blisibimod concentration was 3 µg/mL (47 nM) or higher. No consistent effects on T cells, NK cells, plasmablasts, or plasma cells were evident. The Phase 2 study of blisibimod in subjects with moderate-to-severe SLE (Furie, 2014) demonstrated significantly higher SLE Responder Index (SRI-5) responses in subjects randomized to the highest dose of blisibimod 200 mg once-weekly (QW at Week 20 (p=0.02)).

2.4.4 Clinical Studies With AMG 570

2.4.4.1 FIH Study 20140322

In the ongoing FIH study in healthy volunteers, 7 dose cohorts, over a dose range of 7 mg to 700 mg, are planned with each receiving a single subcutaneous dose of AMG 570 or placebo. Each dose cohort is comprised of 8 subjects with 6 subjects receiving AMG 570 and 2 subjects receiving placebo. As of 1 June 2018, 56 subjects have received a single dose of AMG 570 or placebo at the following doses: 7, 21, 70, 140, 210, 420, or 700 mg SC. 44 subjects have completed the study, 1 subject withdrew consent, 3 subjects have been lost to follow up, and 8 subjects are currently enrolled. In general, AMG 570 was well tolerated at the doses tested. As of 1 June 2018 no deaths or drug-related serious adverse events, dose limiting toxicities or withdrawals due to adverse events (AEs) have been reported.

The FIH study included two humoral immune status stopping rules that accounted for the potential of AMG 570 to decrease total circulating B cells and serum IgG levels. If a single subject in the FIH study satisfied the humoral immune status stopping rules, further studies with AMG 570 would shift from healthy volunteers to patients.

As of the cohort 6 Dose Level Review Meeting on 17 November 2017, no subjects met the humoral immune status stopping rules. Additionally, no clinically significant abnormalities in electrocardiograms (ECGs), safety laboratories, or vital signs have been reported. As of 17 November 2017, blinded safety data from the first 6 cohorts (dose 7, 21, 70, 140, 210 and 420 mg SC) were reviewed by the Dose Level Review team (Amgen in collaboration with study investigators) resulting in a 'safe to proceed to the next dose level' vote. As of 1 June 2018, 8 subjects have been dosed in cohort 7 at the 700 mg SC dose level (allocated 6 subjects AMG 570: 2 subject Placebo).

Additional accumulating safety data from the ongoing 20140322 study will be communicated to investigators during the Dose Level Review meetings for this protocol.

2.4.4.2 FIH Study 20140322 – Clinical Pharmacokinetics

The pharmacokinetics of AMG 570 are currently being assessed in the ongoing FIH study.

Single dose administration of AMG 570 to healthy subjects with subcutaneous (SC) doses of 7, 21, 70, 140, 210, 420 mg demonstrated nonlinear PK and a trend to linear PK at dose higher than 210 mg. This is based on preliminary assessment of dose normalized AUC and C_{max} and PK modeling of the available data.

2.4.4.3 FIH Study 20140322 – Clinical Pharmacodynamics

A validated, semi-quantitative flow cytometry assay was developed to monitor the pharmacodynamics properties of the constituent arms of the bispecific in whole blood specimens.

ICOSL arm – peripheral blood receptor occupancy

ICOSL receptor occupancy was measured using two antibodies: Free ICOSL levels are measured as the MFI of an AMG 570-competing anti-ICOSL antibody conjugated to BV421, gated on circulating B cells. Total ICOSL levels are measured as the MFI of an AMG 570-non-competing anti-ICOSL antibody conjugated to BV421, gated on circulating B cells.

The receptor occupancy (RO) equation is:

$$RO = [1 - (\text{background subtracted free ICOSL} / \text{background subtracted total ICOSL}) / (\text{Baseline background subtracted free ICOSL} / \text{Baseline background subtracted total ICOSL})] * 100$$

Where:

- background subtracted free ICOSL= free ICOSL – Maximally saturated free ICOSL
- background subtracted total ICOSL= total ICOSL – 191

RO is 0 at baseline.

In the ongoing study, preliminary assessment of the pharmacodynamic effects of single doses of AMG 570 indicate dose-related and reversible RO. ICOSL receptor occupancy was measured with two antibodies using a similar approach to published literature ([Sullivan, et al 2016](#)). In the ongoing study, single doses of AMG 570 indicated dose-dependent and reversible RO.

BAFF arm – circulating B cell populations

The effects of BAFF blockade on total circulating B cells as well as naïve and memory B cell subsets (where naïve = CD19+IgD+CD27- and memory = CD19+IgD-CD27+) in the peripheral blood is also being measured.

As of 01 January 2018, preliminary analysis of results from the 6 cohorts of study 20140322, AMG 570 led to a reduction in circulating naïve B cells and an increase in circulating memory B cells compared to the accumulated placebo group. There was an increase in total B cells at day 8, consistent with pharmacodynamic effects of previous BAFF inhibitors ([Witcher, 2016](#) and [Furie, 2014](#)).

2.4.5 Human Exposure Margins

The AMG 570 exposures for this AMG 570 MAD clinical study were based on analysis of observed AMG 570 human PK data from this study or the ongoing AMG 570 FIH study (20140322).

The exposure margins were calculated from serum AMG 570 exposures at the NOAEL of 200 mg/kg in cynomolgus monkeys determined in a 3-month GLP toxicology study, and from the projected human PK exposures after the last dose for the proposed dose levels ([Table 1](#)).

At the NOAEL in monkey, the mean C_{max} following 200 mg/kg dose was 2540 µg/mL and estimated AUC of 23000 µg x day/mL for Q2W dosing regimen. Multiple Q2W x 6 administration of AMG 570 in the dose range of 70 mg to 420 mg SC is projected to be safe and well tolerated, with human exposures less than the exposures at the NOAEL. At the highest planned SC dose (420 mg Q2W X 6), the exposures at the NOAEL in monkeys are 42- and 29-fold above the C_{max} and AUC in humans ([Table 1](#)).

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Table 1. Calculated Exposure Margins for Proposed AMG 570 Doses for Planned Multiple Ascending Dose

Dose (mg)	Route	Human Exposure ^a		Exposure Margin ^b	
		C _{max,last dose} (µg/mL)	AUC _{0-tau,last dose} (µg·day/mL)	C _{max} Ratio ^c	AUC Ratio ^d
70 Q2 wk x 6 (Observed)	SC	6.8	57	370	400
140 Q2 wk x 6 (Predicted)	SC	18	230	141	100
210 Q2 wk x 6 (Predicted)	SC	30	400	85	58
420 Q2 wk x 6 (Predicted)	SC	60	800	42	29

Values reported to two significant figures

AUC_{0-tau} = area under the concentration-time curve from time 0 to 14d after last dose; C_{max} = maximum observed concentration after the last dose; NOAEL = no observed adverse effect level.

^a Predicted AUC_{0-tau} and C_{max} after the last dose, tau = 14 days for Q2W dosing.

^b Based on day 85 toxicokinetic results at the NOAEL dose of 200 mg/kg where AUC₀₋₁₄=23000 µg·day/mL, and C_{max} = 2540 µg/mL (Study 117993).

^c C_{max} ratio is the observed C_{max} in monkeys after the last SC dose divided by the predicted C_{max} in humans after multiple dose.

^d AUC ratio is the observed AUC_{0-tau} in monkeys after the last SC dose divided by the predicted AUC_{0-tau} in humans after multiple dose.

2.4.6 Rationale for Dose Selection

The proposed doses for AMG 570 MAD clinical study were selected based on observed and predicted AMG 570 human exposures at steady-state after multiple administration of proposed doses, as well as interim pharmacodynamics results from the ongoing AMG 570 FIH study (20140322), as described in [section 2.4.4.3](#).

Based on interim pharmacodynamics data from the AMG 570 FIH, the predicted typical individual steady-state exposure after the last dose at 70 mg Q2W × 6 is anticipated to provide sub-optimal pharmacodynamics response with predicted 70% ICOSL RO and at least 40% reduction in B cells. Steady-state exposures at the middle dose of 140 mg Q2W × 6 is anticipated to be associated with near maximal effect on ICOSL RO and B-cell depletion. Dose of 210 mg Q2W × 6 is expected to result in higher ICOSL RO, up to 90% RO or more at steady-state and to maintain and/or increase B cell depletion compared to lower doses. The 420 mg Q2W × 6 dose will allow further evaluation of safety, ICOSL RO response and reduction of B-cells. This dose range will provide additional safety, pharmacokinetics, and pharmacodynamics data that will facilitate the understanding of the exposure-response relationship to inform future studies with AMG 570. The degree of B cell depletion at the highest dose is not anticipated to be

greater than prior BAFF inhibitors, which have demonstrated circulating B cell depletion of up to 70% (Furie RA,2014, Wallace, 2009 , and Isenberg, 2015).

The 210 mg Q2W or 420mg Q2W may be revised based on the data received from the earlier cohorts in this study, as well as revised PK/PD predictions (if any) based on emerging data from the SAD or MAD study.

2.4.7 Rationale for Key Aspects of Study Design

Given the known clinical benefits of modulation of T cell and/or B cell function in RA, subjects with RA are to be enrolled in this multiple ascending dose study. The understanding of both targets of AMG 570, ICOSL and BAFF, and previous experience with AMG 557 and blisibimod with their acceptable safety and tolerability profiles justifies testing in RA patients

The multiple ascending dose design requires thorough evaluation of the safety profile for dose escalation. Subjects will be carefully monitored for adverse events and safety laboratory results. Testing in this population will provide further information on the PK of AMG 570. Furthermore, data from the FIH study will be incorporated into dose escalation decisions to ensure that any dose to be tested at multiple doses has already demonstrated an acceptable safety and tolerability profile in the FIH.

To investigate potential clinical benefit, serologic and clinical markers of disease activity will be investigated. These will include measurement of changes in: DAS28-CRP scores, Patient Global Assessment of Disease Activity, Physician Global Assessment of Disease Activity, CRP, ESR, and autoantibodies associated with RA.

Biomarkers include ICOSL target occupancy on peripheral blood B cells and measurement of serum BAFF concentration. The percentage and absolute counts of total lymphocytes, monocytes, granulocytes and lymphocyte populations including CD3+ T cells, CD16/56+ NK cells, CD19+ B cells as well as naïve and memory B cell subsets (naïve = IgD⁺CD27⁻; memory = IgD⁻CD27⁺) will be determined. The levels of serum IgG and IgM immunoglobulins and BAFF levels will also be measured. Changes in circulating plasma cells will be examined by gene signature, and changes in Tfh cells will be examined by both flow cytometry and gene signature.

2.5 Clinical Hypotheses

Multiple SC dose administration of AMG 570 will achieve acceptable safety and tolerability profiles in RA patients within the proposed dose range (70 to 420 mg SC AMG 570).

3. EXPERIMENTAL PLAN

3.1 Study Design

This is a randomized, placebo-controlled, double-blind, MAD study in subjects with RA. The study consists of 4 SC cohorts. Subjects will be randomized in a 3:1 ratio to receive AMG 570 or placebo according to [Table 2](#).

Table 2. Dose Levels

Cohort #	Planned Dose (mg)	Route	Dosing interval	Number of planned doses	N(active:placebo)
1	70	SC	Q 2 weeks	6	8 (6:2)
2	140	SC	Q 2 weeks	6	8 (6:2)
3	210	SC	Q 2 weeks	6	8 (6:2)
4	420	SC	Q 2 weeks	6	8 (6:2)

The overall study design is described by a [study schema](#) at the end of the protocol synopsis section. The study endpoints are defined in [Section 10.1.1](#).

3.2 Number of Sites

Approximately 8 sites in the U.S. and Germany will be utilized in this study. Additional sites may be added at the discretion of the medical monitor.

3.3 Number of Subjects

Participants in this clinical investigation shall be referred to as “subjects”.

Approximately 32 subjects will be enrolled into 4 cohorts (6 active: 2 placebo in each cohort). See [Section 10.2](#) for sample size rationale.

3.4 Replacement of Subjects

Subjects who are withdrawn or removed from treatment or the study prior to receiving AMG 570 will be replaced at the discretion of the Amgen Medical Monitor and Principal Investigator by notifying the unblinded study pharmacist or designee. The new subject will receive the identical treatment as the replaced subject, but will be assigned a replacement number associated with this new record. The unblinded study pharmacist or designee will retain the randomization list.

3.5 Estimated Study Duration

3.5.1 Study Duration for Subjects

The planned length of participation in the study will be approximately 10 months (22-26 visits) which includes up to 6 weeks for screening, 3-8 days for patient visit,

outpatient visits up to 10 weeks for drug dosing, 24 weeks for evaluation of drug distribution after last dose, followed by end of study (EOS) visit. Subject's participation in the study will conclude at the completion of the end of study procedures. Once complete, this will conclude the subject's participation in the study. However, if an end of study test result demonstrates a significant clinical or laboratory abnormality, the subject will be followed until resolution of the abnormality or until it is considered clinically stable by the Principal Investigator. Specifically, if a subject's CD3-CD19⁺ B cell count is ≥ 107 cells/ μ L at screening and is < 107 cells/ μ L (the lower limit of normal for healthy volunteers (Becton, Dickinson, 2015) at the end of study visit, the subject will return for further testing of B cell counts until the subject has B cell counts ≥ 107 cells/ μ L. If a subject's CD3-CD19⁺ B cell count at screening is between 70-106 (inclusive) at screening and is less than their screening value at the end of study visit, the subject will return for further testing of B cell counts until values are greater than or equal to baseline values. Similarly, if a subject's serum IgG is < 639 mg/dl, the subject will also have serum IgG monitored until the subject has a serum IgG level ≥ 639 mg/dl (Agarwal, 2007). Follow-up visits for B cell counts and IgG levels will continue every 3 months, for up to 12 months after the end of study.

Subject participation may be adjusted based on treatment-emergent data. All adjustments or modifications to the schedule outline above will be agreed upon by the Investigator in consultation with the Sponsor. The Institutional Review Board (IRB) will be informed via written correspondence.

3.5.2 End of Study

Primary Completion: The primary completion date is defined as the date when the last subject is assessed or receives an intervention for the final collection of data for the primary endpoint(s), for the purposes of conducting the primary analysis, whether the study concluded as planned in the protocol or was terminated early.

The primary completion date is the date when the last subject has completed the assessments for EOS visit.

If the study concludes prior to the primary completion date originally planned in the protocol (ie, early termination of the study), then the primary completion will be the date when the last subject is assessed or receives an intervention for evaluation in the study (ie, last subject last visit).

End of Study: The end of study date is defined as the date when the last subject is assessed or receives an intervention for evaluation in the study (ie, last subject last visit), following any additional parts in the study (eg, long-term follow-up for significant clinical or laboratory abnormality), as applicable.

The EOS for each cohort may be prolonged pending treatment-emergent data.

4. SUBJECT ELIGIBILITY

Investigators will be expected to maintain a screening log of all potential study candidates that includes limited information about the potential candidate (eg, date of screening).

Before any study-specific activities/procedure, the appropriate written informed consent must be obtained (see [Section 11.1](#)).

4.1 Inclusion Criteria

101. Subject has provided informed consent prior to initiation of any study-specific activities/procedures.
102. Men or women between 18 to 75 years of age (inclusive) at the time of screening
103. A body mass index (BMI) $\geq 18 \text{ kg/m}^2$ and $\leq 35 \text{ kg/m}^2$
104. Diagnosed with RA as determined by meeting the 2010 ACR/EULAR classification criteria. The duration of disease must be for at least 6 months
105. Disease Activity Score of 28 joints-CRP (DAS28-CRP) > 2.6
106. On a stable dose of methotrexate (MTX) at 5-25 mg weekly for ≥ 4 weeks at day -1. Subjects on MTX must also take folic acid supplementation, as instructed by the investigator.
107. Subjects may continue to take oral corticosteroids, NSAIDs, hydroxychloroquine, or sulfasalazine in combination with MTX provided that doses have remained stable for ≥ 4 weeks prior to day 1 and adhere to the following limits: $\leq 10 \text{ mg/day}$ of prednisone or equivalent, $\leq 400 \text{ mg/day}$ hydroxychloroquine, and $\leq 3000 \text{ mg/day}$ sulfasalazine.
108. Immunizations up to date, as determined by the principal investigator, with a minimum of tetanus (either Td or Tdap), diphtheria, pertussis, seasonal influenza virus (during flu season), and pneumococcal (polysaccharide) vaccinations. For these vaccines, up to date status should be determined by referencing the recommendations for immunizations by the Centers for Disease Control or by equivalent governmental agencies for sites outside the US.
109. Subject (male or female) must be willing to use highly effective contraception during treatment and up to end-of-study. Exceptions to the use of effective contraception includes females of non-reproductive potential, as defined by:
 - History of hysterectomy; OR history of bilateral salpingectomy; OR history of bilateral oophorectomy; OR
 - Female subject is considered to be postmenopausal due to absence of vaginal bleeding or spotting for at least 12 months and:

- If less than 55 years of age, follicle stimulating hormone (FSH) and estradiol must be within the laboratory's reference range for postmenopausal females.
- If 55 to 59 years of age and there is uncertainty regarding menopausal status, FSH must be within the laboratory's reference range for postmenopausal females
- If 60 years of age or older, evaluation of FSH is not needed to confirm postmenopausal status

4.2 Exclusion Criteria

201. Subject likely to not be available to complete protocol-required study visits or procedures, and/or to comply with all required study procedures to the best of the subject and investigator's knowledge.
202. History or evidence of any other clinically significant disorder, condition or disease that, in the opinion of the investigator or Amgen physician, would pose a risk to subject safety or interfere with the study evaluation, procedures or completion.
203. Subjects who fulfill any of the following criteria are not eligible for the study:
 - Men and women of reproductive potential who are unwilling to practice an acceptable method(s) of effective birth control while on study. Acceptable methods of effective birth control include sexual abstinence (men, women); vasectomy; bilateral tubal ligation/occlusion; or a condom with spermicide (men) in combination with barrier methods (diaphragm, cervical cap or cervical sponge), hormonal birth control or IUD (women).
 - Women who are lactating/breastfeeding or who plan to breastfeed while on study.
 - Women with a positive pregnancy test.
 - Women planning to become pregnant while on study.
 - Men with partners who are pregnant or planning to become pregnant while the subject is on study.
 - Men who are unwilling to abstain from sperm donation while on study.
204. Uncontrolled, clinically significant systemic disease other than RA such as diabetes mellitus, liver disease, asthma, cardiovascular disease or hypertension
205. Malignancy within 5 years (except successfully treated in situ cervical cancer or squamous or basal cell carcinoma of the skin)
206. Presence of a serious infection (defined as requiring hospitalization or IV antibiotics within 8 weeks before Screening), recurrent/chronic infections (defined as ≥ 3 infections requiring antibiotics over the past 12 months prior to Screening), Prosthetic joint infection within 5 years of screening, or native joint infection within 1 year of screening
207. Class IV RA ([Hochberg et al, 1992](#)) according to ACR revised response criteria
208. Diagnosed with Felty's syndrome (RA, splenomegaly, and granulocytopenia)
209. Known or suspected sensitivity to mammalian cell-derived (ie, from Chinese hamster ovary) products or any components of the study drug

210. History of alcohol and/or substance abuse within the last 12 months
211. Chronic pain syndrome requiring daily narcotic analgesics or morphine derived medication for analgesic relief at screening
212. All herbal medicines (eg, St. John's wort), vitamins, and supplements consumed by the subject within the 30 days prior to receiving the first dose of investigational product, and continuing use if applicable, that has not been reviewed by the investigator, in consultation with the Amgen medical monitor, if needed.
213. Subjects who have received ≥ 80 mg of prednisolone equivalents via intra-articular or systemic corticosteroid injections for treatment of acute RA flare (not being part of a regular therapeutic regimen) within 4 weeks prior to randomization
214. Use of cyclophosphamide, cyclosporine, mycophenolic acid, or gold within the previous 6 months prior to day -1
215. The use of any commercial parenteral biologic or oral synthetic DMARD (such as tofacitinib or baracitinib) in the 3 months prior to enrollment with the exception of rituximab (see below)
216. Receipt of rituximab at any time in the past
217. Subjects who fulfill any of the following criteria are not eligible for the study:
 - Receiving or has received any investigational drug (or is currently using an investigational device) within the 30 days or 5 half-lives (whichever is longer), prior to receiving the first dose of AMG 570.
 - Subjects who were previously exposed to AMG 570
218. The use of any other experimental/investigational biologic DMARD that impacts the immune system requires a) > 6 months off the investigational agent prior to enrollment, b) approval by the sponsor, and c) reasonable confidence by the investigator that the pharmacodynamic effect of the agent has returned to baseline at the time of enrollment.
219. Subject received a live attenuated vaccine within 3 months before screening or a non-live, inactivated vaccine within 28 days of randomization or plan to receive live attenuated vaccine within 30 days and 5 half-lives of the last dose of study medication
220. Subjects with a history of residential exposure (within 6 months) to tuberculosis (TB) without a documented history of prophylactic treatment or subject has a positive interferon-gamma release assay (QuantiFERON® Gold or equivalent) within 6 weeks before randomization
221. Subject is positive for HIV antibodies, hepatitis B surface antigen, hepatitis B core antibody (confirmed by hepatitis B deoxyribonucleic acid [DNA] polymerase chain reaction [PCR] test) or detectable hepatitis C virus ribonucleic acid [RNA] by PCR (screening is generally done by hepatitis C antibody [HepCAb], followed by hepatitis C virus RNA by PCR if HepCAb is positive). Subjects with a history of hepatitis B vaccination without history of hepatitis B are allowed.
222. Clinically significant abnormalities on ECG (as defined by the Investigator) and/or QTc > 480 msec.

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- 223. CD3-CD19+ B cell count < 70 cells/ μ L. For B cell counts of 70-106, a discussion with the Amgen medical monitor to include evaluation of humoral immune functional status (assessed by IgM and IgG levels as well as history of recurrent infections) is required for pre-approval prior to enrollment
- 224. Serum IgM or IgG level < LLN (measured at a central lab)
- 225. White blood cell count < $3.0 \times 10^3/\mu$ L, or absolute neutrophil count (ANC) < $2.0 \times 10^3/\mu$ L or platelet count < $140 \times 10^9/L$
- 226. Elevated AST or ALT > 1.5 x upper limit of normal at Screening if subjects are receiving MTX.
- 227. Elevated serum creatinine > 1.5 x upper limit of normal at screening
- 228. Positive urine screen for alcohol and/or potential drugs of abuse at screening or Day -1
- 229. Blood donation of > 500 mL within 60 days of study drug administration
- 230. Evidence of renal disease, defined as: Calculated glomerular filtration rate < 60 mL/min using the Modification of Diet in Renal Diseases (MDRD) equation: $186 \times (\text{Creatinine}/88.4)^{-1.154} \times (\text{Age})^{-0.203} \times (0.742 \text{ if female}) \times (1.210 \text{ if black})$

5. SUBJECT ENROLLMENT

Before subjects begin participation in any study-specific activities/procedures, Amgen requires a copy of the site's written institutional review board/independent ethics committee (IRB/IEC) approval of the protocol, informed consent form, and all other subject information and/or recruitment material, if applicable (see [Section 11.2](#)). All subjects must personally sign and date the informed consent form before commencement of study-specific activities/procedures. Adverse Events are to be collected for a subject once they are enrolled in the study. A subject is considered enrolled when the investigator decides that the subject has met all eligibility criteria.

The Investigator is to document the enrollment decision and date, in the subject's medical record and in/on the enrollment CRF.

Each subject who enters into the screening period for the study, which is the first time that a subject undergoes detailed assessments to determine their potential for eligibility, receives a unique subject identification number before any study procedures are performed. The subject identification number will be assigned manually. This number will be used to identify the subject throughout the clinical study and must be used on all study documentation related to that subject. A screen failed subject may be rescreened up to two times at the discretion of the PI with the agreement of the Amgen Medical Monitor.

The subject identification number must remain constant throughout the entire clinical study; it must not be changed after initial assignment, including if a subject is rescreened. This number will not necessarily be the same as the randomization number assigned for the study.

5.1 Randomization/Treatment Assignment

Randomization to AMG 570 or placebo will be based on a randomization schedule prepared by Amgen before the start of the study and retained by the unblinded study pharmacist or designee. After eligibility for study participation has been confirmed (based on data collected during screening visits and on Day -1), eligible subjects will be randomized to receive either AMG 570 or placebo. The randomization date is to be documented in the subject's medical record and on the enrollment eCRF.

Unique randomization numbers will be assigned in sequential order according to when each subject met the eligibility criteria. At no time will the same randomization number be assigned to more than one subject.

The randomization date is to be documented in the subject's medical record and on the enrollment case report form (CRF).

In the event a subject is randomized, but withdraws before receiving study medication, or during the study for reasons other than adverse events ([Section 8.3](#)), a replacement subject may be enrolled at the discretion of the Amgen Medical Monitor and Principal Investigator. The unblinded study pharmacist or designee will be notified. The new subject will receive the identical treatment as the replaced subject but will be assigned a replacement number per the randomization list.

5.2 Site Personnel Access to Individual Treatment Assignments

A subject's treatment assignment should only be unblinded at study site when knowledge of the treatment is essential for the further management of the subject on this study or may potentially impact the safety of subjects currently enrolled or subjects in subsequent cohorts. Unblinding at the study site for any other reason will be considered a protocol deviation. The investigator is strongly encouraged to contact the Amgen Study Manager before unblinding any subject's treatment assignment, but must do so within 1 working day after the event.

Treatment assignments will be unblinded at study site after database lock. After initial database lock and receipt of written authorization from Amgen to unblind, the unblinded pharmacist will release the specified unblinded pharmacy records to site staff designated

to enter the subject treatment into each subject's Unblinded Investigational Product Administration Case Report Form (CRF).

6. TREATMENT PROCEDURES

The planned doses of AMG 570 are 70 mg SC, 140 mg SC, 210 mg SC and 420 mg SC administered once every 2 weeks for 10 weeks (last study drug administration on Day 71/Week 10). Each randomized subject will receive a total of 6 SC doses of AMG 570 or matching placebo. Dose escalation review meetings will be held prior to escalation to a higher dose. Dose escalation will only occur provided adequate safety from the prior multiple dose cohort, and no dose will be given unless adequate safety from the equivalent dose tested in the single ascending dose study (20140322) with AMG 570 has also been determined to have adequate safety.

6.1 Classification of Product(s), Medical Device(s), and/or Combination Product(s)

The Amgen Investigational Product and/or placebo used in this study include: AMG 570 and placebo. The Investigational Product Instruction Manual (IPIM), a document external to this protocol, contains detailed information regarding the storage, preparation, destruction, and administration of AMG 570 and placebo.

6.2 Investigational Product

All investigational products will be administered at the research facility by a qualified staff member. A physician must be present at the time of investigational product administration.

6.2.1 Amgen Investigational Product AMG 570

AMG 570 will be manufactured and packaged by Amgen Inc. and distributed using Amgen clinical study drug distribution procedures. AMG 570, which is a clear colorless sterile solution which will be packaged in open label 5mL glass vials containing 70 mg/ml of AMG 570 formulated with 10mM acetate, 9.0%(w/v) sucrose, 0.01%(w/v) polysorbate 80 at pH 5.2. Placebo will be presented in identical containers and stored/package the same as AMG 570, but will not contain AMG 570 protein. All AMG 570 and placebo will be shipped to the study site and should be stored the same as AMG 570 at 2°C to 8°C with limited exposure to light.

For more information regarding investigational product handling and preparation please see the study specific IPIM which is provided as a separate document.

6.2.1.1 Dosage, Administration, and Schedule

All study drug will be administered (Table 2) at the research facility by a qualified staff member. A physician must be available during each administration of investigational product. The date, time, lot number and volume of investigational product administered will be recorded on the individual subject's electronic case report form (eCRF). Subject will be randomized only once to one of 4 Cohorts starting with Cohort 1 and continuing in a sequential manner to Cohort 4. Within each cohort, subjects will be randomized to receive 6 SC doses of AMG 570 or placebo in a ratio of 6:2. Once a dose is selected for a cohort, no dose adjustments will be made for an individual subject within the cohort.

Subcutaneous Administration of AMG 570/Placebo

The appropriate volume of the IP administered will be based on the set dosage amounts. Doses for Cohorts 1-3 will be administered as a 1 mL per injection format (up to 3 x 1mL injections depending on treatment cohort). For Cohort 4, doses will be administered as a 1.5 mL per injection format (4 x 1.5 mL). For individual SC dose volumes that exceed 1.0 mL, the dose will be split into separate syringes and administered into different sites on the subject's anterior abdominal wall. Using the umbilicus as a center point, the abdomen will be divided into four quadrants – the right upper quadrant (RUQ), left upper quadrant (LUQ), right lower quadrant (RLQ) and left lower quadrant (LLQ). Each quadrant may be injected once, starting with the subjects' upper right, and progressing in a clockwise manner to the upper left, lower left and lower right quadrants.

6.2.1.2 Dose-cohort Study Escalation and Stopping Rules

Dose level review meetings (DLRM) will be held to review data, monitor safety, and make dosing decisions. The DLRM members will be composed of the investigator(s), Amgen Medical Monitor, Amgen Global Safety Officer (GSO) or designee, Early Development Leader or designee, Clinical Study Manager or designee, and Biostatistics representative or designee. Additional members may be added as needed (eg, PK Scientist). The DLRM voting members will include the investigator(s), Amgen Medical Monitor and Amgen GSO or designee.

The DLRM voting members will be responsible for dosing decisions, which may include:

- 1) escalation to the next planned dose
- 2) escalation to an intermediate dose (a dose lower than the next planned dose)
- 3) de-escalation to a lower dose
- 4) continuation, delay, or termination of dosing
- 5) repetition or expansion of a cohort.

Dose escalation for all cohorts will be based upon unanimous decision of the DLRM members.

The dosing decisions will be based on the review of safety data during the DLRM. Within each cohort, the safety data will be assessed after all 8 subjects have been enrolled and at least 6 subjects have completed the day 29 visit, which corresponds to receipt of the third dose of IP. In addition to the review of safety data, emerging PK and PD data will also be reviewed in a blinded fashion at the DLRM as these data become available. Emerging safety, PK, and PD data from equivalent dose level cohorts of the AMG 570 Single Ascending Dose (SAD) study in healthy volunteers will also be reviewed prior to dose escalation in the multiple ascending dose study. As of 1 June 2018, blinded safety review for the 700 mg SC dose has not yet occurred for the SAD. Thus, dose escalation for cohort 4 will only proceed if safety, PK, and PD data from the 700 mg SC cohort of the SAD have been reviewed by the DLRM and no significant safety concerns are observed.

Dose adjustments (if any) will be made on a treatment cohort basis and not on an individual basis, and will be agreed upon by Amgen after reviewing emerging safety, PK, and/or PD data.

The , 210 mg Q2W or 420 mg Q2W doses may be revised based on the data received from the earlier cohorts in this study, the higher doses evaluated in the ongoing FIH SAD study in healthy subjects, as well as revised PK/PD predictions (if any) based on emerging data from the SAD or MAD study.

Dose Stopping and Review

Further dosing of AMG 570 will be either stopped or modified to a lower dose if suspected adverse drug reactions and/or changes in safety data (including but not limited to vital signs, ECGs, clinical laboratory results, or laboratory parameters reflecting humoral immune status) are observed and these changes pose a health risk. A DLRM is held when a dose limiting toxicity (DLT) has occurred. A DLT is defined as any treatment-related fatal, life threatening or disabling serious adverse event (SAE). In addition, any adverse event or change in vital signs, clinical laboratory test value, or ECG which is considered drug related, and poses a significant health risk, as determined by the Investigators and Sponsor, would also constitute a DLT and form the basis for stopping dose escalation in the study. In addition, dosing will be stopped or modified if any of the scenarios shown in [Table 3](#) are met.

Dose Cohort Stopping Rules

This section addresses how subject safety will be monitored through adherence to rules for stopping dosing within a dose cohort (which precludes any further dose escalation).

[Table 3](#) describes the dose stopping rules to be used for stopping dosing within a cohort.

Table 3. Dose Cohort Stopping Rules

Scenario	Action
Occurrence of a CTCAE Grade 3 or greater adverse event in at least 2 subjects in a dose cohort	Stop dosing additional subjects in the cohort and convene DLRM.
OR	Review AE and all relevant safety data for evidence of relationship to treatment and clinical significance. Consider unblinding to determine relatedness to investigational product.*
Any adverse event or change in above safety parameters, which is considered drug related, and poses a significant health risk to the subject.	If Grade 3 or higher adverse event is determined to be related to study drug, and clinically significant by the DLRM, no further dose escalation to proceed.
OR	Otherwise, upon unanimous decision of the review team, one of the following decisions may be made:
A change in vital signs, clinical laboratory value, or ECG that would be classified as Grade 3 or higher by the CTCAE v. 4.0 and considered drug related	• Enrollment of the cohort may resume, • The cohort may be expanded at the same dose, • A lower dose cohort may be added to the study.
OR	
A fatal, life threatening or disabling serious adverse event (SAE)	

* A subject's treatment assignment should only be unblinded when knowledge of the treatment is essential for the further management of the subject, or may impact the safety of subjects currently enrolled, or subjects in subsequent cohorts

Dosing will be stopped or modified by the DLRM members if treatment-related AEs, changes in vital signs, ECGs, or clinical laboratory results are observed and these changes pose a significant health risk.

The dosing schedule is described by a [schema](#) in the protocol synopsis.

6.2.1.3 Dosage Adjustments, Delays, Rules for Withholding or Restarting, Permanent Discontinuation

This protocol allows for some alterations from the currently outlined dosing schedule, but the maximum dose will not exceed 420 mg.

The actual dose to be administered may be adjusted by the DLRM voting members based on safety, tolerability, and preliminary PK data of previous dose levels. Dose adjustments may involve either an increase or a decrease in the planned dose, but will not exceed pharmacokinetic criteria defined in the Rationale for Dose Selection ([Section 2.4.6](#)).

If consistent adverse events of moderate or severe intensity occur in the group, or if unacceptable pharmacological effects, reasonably attributable in the opinion of the investigator to dosing with AMG 570, are observed in more than 2 subjects in a group, the dose escalation will be temporarily halted and no additional subjects will be dosed until a full safety review by DLRM of the study has taken place. Relevant data will be reviewed and discussed with the investigator, appropriate Amgen personnel, and IRB prior to resumption of dosing.

Clinically or medically significant suspected adverse drug reactions, and serious adverse events considered to be related to study procedures will be followed until resolved or considered stable. The study may be terminated at any point in time at the discretion of the sponsor.

6.3 Other Protocol-required Therapies

All subjects must maintain a stable dose and the same route of administration of MTX. The doses of MTX should be taken on the same day of the week. Subjects must also take folic acid supplementation, as instructed by the investigator, during the trial if receiving MTX. Dose adjustment of MTX will be limited as follows:

- Subjects who have documented toxicity of MTX (defined as ALT/AST > 2x ULN for 2 consecutive measures within 14 days) may have 2 dose reductions of up to 5 mg/week each. However, for those subjects with < 15 mg/week dose at study entry, dose reduction of 2.5 mg/week will be allowed for 2 weeks and then will be returned to full dose. If the toxicity episode persists, the MTX dose can be reduced to the next level at the discretion of the investigator.
- Subjects experiencing a flare in the underlying RA during the study treatment period may receive a single 7 day course of oral corticosteroids as described below.

6.4 Concomitant Therapy

Throughout the study, investigators may prescribe any concomitant medications or treatments deemed necessary to provide adequate supportive care. All use of concomitant medications and supplements (nutritional and prophylactic) will be recorded on the applicable eCRF.

After the completion of Week 13 visit procedures, subjects may receive additional therapeutic interventions to treat signs and symptoms of RA and to enable the subject to remain on study for follow-up assessments through Week End of Study (EOS) as indicated in the Schedules of Assessments ([Section 7.1](#)).

DMARDS

New medications to treat RA including biologic and non-biologic DMARDS must not be introduced during the study unless it is deemed necessary to provide adequate supportive care for any intercurrent illness. Any subject who receives a new concomitant DMARD for either RA or another concurrent illness during the course of the study should be discontinued from investigational product, and the appropriate evaluations and case report form (CRF) must be completed. Unless the subject withdraws full consent and early terminates from the study, the subject will be requested to return for follow up visits.

NSAIDs

Subjects taking NSAIDs must be on stable dose $\geq$ 4 weeks prior to screening and remain on stable dose during the treatment period of the study. If necessary for safety reasons, the dose of NSAIDs can be reduced or stopped during the duration of the study. In cases of flare, the dose of NSAIDs can be temporarily increased as needed. However, the subject must return to the maintenance dose (the dose at baseline) as soon as the flare resolves. This includes the addition of an NSAID to treat flare in subjects who were not taking an NSAID at baseline.

Hydroxychloroquine

Subjects may remain on a stable dose of hydroxychloroquine up to 400 mg/day in combination with DMARDs.

Sulfasalazine

Subjects may remain on a stable dose of sulfasalazine up to 3000 mg/day in combination with DMARDs.

Analgesics

Rescue analgesics (eg, hydrocodone, oxycodone, codeine, propoxyphene, and/or dextropropoxyphene) may be used for up to 5 consecutive days except within 12 hours before a scheduled study efficacy evaluation.

Corticosteroids

Subjects taking oral corticosteroids (not to exceed the equivalent of 10 mg of Prednisone per day) must be on stable dose $\geq$ 4 weeks prior to screening and remain on stable dose during the treatment period of the study. In the event of worsening RA, subjects may receive a single 7 day course of oral corticosteroids following approval by Amgen. This

rescue regimen of corticosteroids (up to 10 mg/day of Prednisone above their stable baseline dose) must return to baseline levels within 2 weeks.

Joint count assessments for 2 weeks following the receipt of the increased corticosteroids will be excluded from efficacy analyses. Intra-articular corticosteroid injections are not permitted during the treatment period of the study (through week 12). An increase in the dose of corticosteroids due to elective surgery is permitted. Should a subject require an increase in his/her corticosteroid dose during the peri-operative stage, every attempt should be made to reduce the corticosteroid dose to their baseline dose within 14 days post surgery. Joint count assessments for 2 weeks following the receipt of the increased corticosteroids will be excluded from efficacy analyses.

6.5 Hepatotoxicity Stopping Rules

6.5.1 Criteria for Permanent Withholding of AMG 570 due to Potential Hepatotoxicity

AMG 570 should be permanently withheld, and the subjects should be followed according to the recommendations in [Appendix A](#) (Additional Safety Assessment Information) for possible drug-induced liver injury (DILI), if ALL of the criteria below are met:

- Total Bilirubin > 2x upper limit of normal (ULN) or INR > 1.5

AND

- ALT or AST >3x ULN

AND

- No other cause for the combination of the above laboratory abnormalities is immediately apparent; important alternative causes for elevated AST/ALT and/or elevated TBL values include, but are not limited to:
 - Hepatobiliary tract disease
 - Viral hepatitis (eg, hepatitis A/B/C/D/E, Epstein-Barr virus, cytomegalovirus, herpes simplex virus, varicella, toxoplasmosis, and parvovirus)
 - Right sided heart failure, hypotension, or any cause of hypoxia to the liver causing ischemia
 - Exposure to hepatotoxic agents/drugs or hepatotoxins including herbal and dietary supplements, plants, and mushrooms,
 - Heritable disorders causing impaired glucuronidation (eg, Gilbert's syndrome, Crigler-Najjar syndrome) and drugs that inhibit bilirubin glucuronidation (eg, indinavir, atazanavir)
 - Alpha-one antitrypsin deficiency
 - Alcoholic hepatitis

- Autoimmune hepatitis
- Wilson's disease and hemochromatosis
- Nonalcoholic fatty liver disease including steatohepatitis
- Non-hepatic causes (eg, rhabdomyolysis, hemolysis)

6.6 Alcohol, Tobacco, and Exercise Restrictions

- Prior to admission to the clinic for each clinic visit, subject will abstain from alcohol consumption for 24 hours.
- Smoking will be limited to no more than 10 cigarettes per day
- Subjects are required to refrain from strenuous exercise 48 hours prior to each blood collection for clinical laboratory testing. Subjects may participate in light recreational activities during the study.

6.7 Product Complaints

A product complaint is any written, electronic or oral communication that alleges deficiencies related to the identity, quality, durability, reliability, safety, effectiveness, or performance of a drug(s) or device(s) after it is released for distribution to market or clinic by either Amgen or by distributors and partners for whom Amgen manufactures the material.

This includes any drug(s), device(s), or combination product(s) provisioned and/or repackaged /modified by Amgen. Drug(s) or device(s) includes investigational product.

Any product complaint(s) associated with an investigational product(s) or non-investigational product(s) or device(s) supplied by Amgen are to be reported according to the instructions provided in the IPIM.

7. STUDY PROCEDURES

7.1 Schedule of Assessments

See the following pages for Schedule of Assessments.

Approved

Table 4. Schedule of Assessments – Screening Through Day 43 (Week 6)

	Screening	Baseline	Treatment														
Week	- 6 to 0		0											1	2	4	6
Study Day ^[1]	- 42 to -1	-1	1 ^[1]							2	3	4	7	8	15	29	43
Relative to FIRST DOSE			Pre	0	0.5 h	1 h	4 h	6 h	12 h	24 h	48 h	72 h	144 h	168 h	336 h	28 d	42 d
Treatment Assignment		x															
Dosing ^[2]				x											x	x	x
General Assessments																	
ICF	x																
Eligibility	x																
Medical/Med/Drug/Alcohol History	x																
Demographics	x																
BMI (Ht. & Wt.)	x																
Clinical Assessments of Disease Activity																	
Tender/Swollen Joint Count (28 joints)	x	x													x	x	
Patient Global Assessment of Disease Activity	x	x													x	x	
Physician Global Assessment of Disease Activity	x	x													x	x	
HAQ-DI	x	x													x	x	
Safety Assessments																	
Physical Exam	x	x									x					x	x
Vitals	x	x	x	x	x	x	x	x	x	x	x			x	x	x	
ECG ^[3]	x	x	x ^[4]	x		x	x		x	x	x	x					
AEs and Con Meds			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Serious Adverse Events	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Lab Assessments																	
Drug/Alcohol Screen	x	x														x	
HIV, HepC, HepB	x	x															
Serum Chemistry	x	x									x				x	x	x
Hematology, including reticulocyte count and CBC w/Differential	x	x									x				x	x	x

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^[1] Inpatient visit (3-8 days) may be scheduled for 1st dose and 6th dose (Day 1 and Day 71)

^[2] For visits when dosing is required, administration of investigational product should occur after other required assessments have been completed.

^[3] ECGs must be done in triplicate and obtained before blood samples are drawn.

^[4] ECG Day 1 pre-dose – Triplicate ECGs will be performed on 3 occasions separated by at least 30 minutes (total 9 ECGs)

Table 4. Schedule of Assessments – Screening Through Day 43 (Week 6)

	Screening	Baseline	Treatment														
Week	- 6 to 0		0											1	2	4	6
Study Day ^[1]	- 42 to -1	-1	1 ^[1]							2	3	4	7	8	15	29	43
Relative to FIRST DOSE			Pre	0	0.5 h	1 h	4 h	6 h	12 h	24 h	48 h	72 h	144 h	168 h	336 h	28 d	42 d
Treatment Assignment		x															
Dosing ^[2]				x											x	x	x
Lab Assessments (continued)																	
Urinalysis	x	x									x				x	x	x
b-HCG	x	x															
Urine Pregnancy Test	x	x															
FSH/Estradiol	x																
C-Reactive Protein (CRP)	x	x														x	
QuantiFERON-TB Gold	x																
Erythrocyte Sedimentation Rate (ESR)	x	x														x	
PK Assessments																	
PK Draw ^[3]			x					x	x	x	x	x	x	x	x	x	x
Biomarker & Immunological Assessments																	
ADA			x														
Immunophenotyping and Receptor Occupancy (IPRO) Assay by flow cytometry	x		x											x	x	x	x
Tfh by flow cytometry			x													x	
Whole blood microarray		x	x													x	
IgG, IgA and IgM Serum	x		x													x	x
BAFF Serum Levels			x													x	
Rheumatoid Factor (RF total and isotypes)			x													x	
ACPAs (αCCP)			x													x	
Exploratory Biomarkers (serum and plasma)			x													x	
Exploratory Autoimmune Antibodies			x													x	
Exploratory Pharmacogenomics cell pellet			x														

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^[1] Inpatient visit (3-8 days) may be scheduled for 1st dose and 6th dose (Day 1 and Day 71)

^[2] For visits when dosing is required, administration of investigational product should occur after other required assessments have been completed.

^[3] When PK draw and IP dosing are scheduled on the same day, PK draw should be collected pre-dose.

Table 5. Schedule of Assessments – Day 57 (Week 8) Through End of Study (Week 34)

	Treatment								Follow-up											EOS
Week	8	10							11	12	13	14	15	16	18	20	22	26	30	34
Study Day ^[1]	57	71 ^[1]			72	73	74	77	78	85	92	99	106	113	127	141	155	183	211	239
Relative to LAST DOSE		0	6h	12 h	24 h	48 h	72 h	144 h	168 h	336 h	3 w	4 w	5 w	6 w	8 w	10 w	12 w	16 w	20 w	24 w
Treatment Assignment																				
Dosing ^[2]	x	x																		
Clinical Assessments of Disease Activity																				
Tender/Swollen Joint Count (28 joints)	x									x				x		x		x		x
Patient Global Assessment of Disease Activity	x									x				x		x		x		x
Physician Global Assessment of Disease Activity	x									x				x		x		x		x
HAQ-DI	x									x				x		x		x		x
Safety Assessments																				
Physical Exam	x	x										x			x		x	x	x	x
Vitals	x	x										x			x		x	x	x	x
ECG ^[3]				x	x	x	x													x
AEs and Con Meds	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Serious Adverse Events	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Lab Assessments																				
Drug/Alcohol Screen	x									x										x
HIV, HepC, HepB																				
Serum Chemistry	x	x								x				x		x		x		x
Hematology, including reticulocyte count and CBC w/ Differential	x	x								x				x		x		x		x
Urinalysis	x	x								x				x		x		x		x
b-HCG																				
Urine Pregnancy Test										x										x
FSH/Estradiol																				

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^[1] Inpatient visit (3-8 days) may be scheduled for 1st dose and 6th dose (Day 1 and Day 71)

^[2] For visits when dosing is required, administration of investigational product should occur after other required assessments have been completed.

^[3] ECGs must be done in triplicate and obtained before blood samples are drawn.

Table 5. Schedule of Assessments – Day 57 (Week 8) Through End of Study (Week 34)

	Treatment								Follow-up											EOS
Week	8	10							11	12	13	14	15	16	18	20	22	26	30	34
Study Day ^[1]	57	71 ^[1]			72	73	74	77	78	85	92	99	106	113	127	141	155	183	211	239
Relative to LAST DOSE		0	6h	12 h	24 h	48 h	72 h	144 h	168 h	336 h	3 w	4 w	5 w	6 w	8 w	10 w	12 w	16 w	20 w	24 w
Treatment Assignment																				
Dosing ^[2]	x	x																		
Lab Assessments (continued)																				
C-Reactive Protein (CRP)	x									x				x		x		x		x
QuantiFERON-TB Gold																				
Erythrocyte Sedimentation Rate (ESR)	x									x				x		x		x		x
PK Assessments																				
PK Draw ^[3]	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x			x
Biomarker & Immunological Assessments																				
ADA	x																x			x
Immunophenotyping and Receptor Occupancy (IPRO) Assay by flow cytometry	x	x							x	x		x			x		x			x ^[4]
Tfh by flow cytometry	x									x							x			x
Whole blood microarray	x									x							x			x
IgG, IgA and IgM Serum	x									x							x			x ^[5]
BAFF Serum Levels	x									x							x			x
Rheumatoid Factor (RFtotal and isotypes)	x									x							x			x
ACPAs (αCCP)	x									x							x			x
Exploratory Biomarkers (serum and plasma)	x									x							x			x
Exploratory Autoimmune Antibodies	x									x							x			x

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^[1] Inpatient visit (3-8 days) may be scheduled for 1st dose and 6th dose (Day 1 and Day 71)

^[2] For visits when dosing is required, administration of investigational product should occur after other required assessments have been completed.

^[3] When PK draw and IP dosing are scheduled on the same day, PK draw should be collected pre-dose.

^[4] B cell repletion: If a subject's CD3-CD19+ B cell counts are < 107 cells/uL (the lower limit of normal for healthy volunteers), the subject will return for further testing of B cell counts until the subject has B cell counts ≥ 107 cells/uL or greater than or equal to their screening B cell value if the screening B cell count was 70-106 cells/uL. Follow-up visits for B cell counts will continue every 3 months, for up to 12 months after the end of study.

^[5] IgG repletion: If a subject's IgG levels are < 639 mg/dL (the lower limit of normal for healthy volunteers), the subject will return for further testing of IgG levels until the subject has IgG levels ≥ 639 mg/dL. Follow-up visits for IgG levels will continue every 3 months, for up to 12 months after the end of study.

Refer to the applicable supplemental laboratory and electrocardiogram [ECG] manuals for detailed collection and handling procedures. Ensure these manual(s) are provided separately from the protocol to the sites.

7.2 General Study Procedures

Adherence to the study design requirements, including those specified in the Schedule of Assessments ([Section 7.1](#)) are essential and required for study conduct.

This section lists the procedure and parameters of each planned study assessment. The exact timing of each assessment is listed in the Schedule of Assessments ([Section 7.1](#)).

Whenever vital signs, 12-lead ECGs, and blood draws are scheduled for the same nominal time, the assessment should occur in the following order: ECGs, vital signs, blood draws. The timing of the assessment should allow the blood draw to occur at the exact nominal time.

The timing and number of planned study assessments, including: Safety, PK, PD and Immunogenicity assessment may be altered during the course of the study based on newly available data (eg, to obtain data closer to the time of peak plasma concentration) to ensure appropriate monitoring.

The IRB will be informed of any safety issues that require alteration of the safety monitoring scheme or amendment of the Informed Consent Form.

No more than 300 mL of blood, including from any extra assessments that may be required, will be collected over any 8-week period for duration of the study.

Additional procedures deemed necessary as part of standard of care or as required by local laws and regulations may be performed at the Investigator's discretion.

7.2.1 Screening

Before any study-related screening or baseline procedure can be completed, a subject must sign and date the IRB approved ICF.

The following procedures are to be completed during the screening period and may also be completed at time points designated in the Schedule of Assessments ([Section 7.1](#)).

- Confirmation that the Informed Consent Form has been signed
- Demographic data including sex, age, race, and ethnicity will be collected in order to study their possible association with subject safety and treatment effectiveness. Additionally demographic data will be used to study the impact pharmacodynamics and pharmacokinetics of the protocol- required therapies.
- Medical History
- Complete Physical Examination
- BMI (Height and Weight)
- Vital signs (blood pressure, heart rate, respiration rate, temperature)
- Electrocardiogram (ECG)
- Clinical Assessments of Disease Activity
 - Tender/Swollen 28-Joint Count
 - Patient Global Assessment of Disease Activity
 - Physician Global Assessment of Disease Activity
 - HAQ-DI
- Drug and Alcohol Screen
- HIV (human immunodeficiency virus), HBcAb (hepatitis B core antibody), HepBsAg (hepatitis B surface antigen), HepCAb (Hepatitis C antibody)
- Laboratory Assessments (including creatinine clearance, chemistry, hematology, and urinalysis)
- b-HCG
- Follicle-Stimulating Hormone and estradiol (postmenopausal females only) and Pregnancy Test (females only)
- C-Reactive Protein (CRP)
- QuantiFERON-TB Gold
- Erythrocyte Sedimentation Rate (ESR)
- Immunophenotyping and Receptor Occupancy, IPRO assay, IgG and IgM
- Serious Adverse Event reporting

Subjects may be rescreened only with permission of the Amgen medical monitor. A new informed consent form must be signed unless it has been < 30 days since the previous ICF signature was obtained ([Section 5](#)).

7.2.2 Day -1 and Treatment

The following procedures will be completed during the treatment period at the times designated in the Schedule of Assessments ([Section 7.1](#)). Administration of protocol-required therapies is to be administered last during each visit that it is required.

Subjects will be admitted to the clinical unit on Day -1 at which time the following assessments will be obtained:

- Physical Examination
- Vital Signs
- ECG
- Clinical Assessments of Disease Activity
 - Tender/Swollen 28-Joint Count
 - Patient Global Assessment of Disease Activity
 - Physician Global Assessment of Disease Activity
 - HAQ-DI
- Drug and Alcohol Screen
- Laboratory Assessments (chemistry, hematology, and urinalysis)
- Laboratory Assessments of Disease Activity (CRP and ESR)
- Pregnancy Testing (females only)
- Treatment Assignment (can occur any time prior to dosing)
- Whole blood microarray
- Serious Adverse Event reporting

Subject will stay overnight and then undergo Day 1 assessments as follows:

- Pre-dose Vitals
- Pre-dose ECGs
- Pre-dose Biomarker and Immunological Assessments
- Pre-dose PK
- Dosing
- Post-dose Vitals
- Post-dose ECG(s)
- Post-dose PK draws
- Adverse Events and Serious Adverse Events reporting
- Documentation of concomitant and rescue medications

Additional dosing days and assessments will be conducted per cohort assignment and dosing schedule.

Subject will return to clinical unit on an outpatient basis for scheduled study procedures including blood draws for PK assessment, safety, immunogenicity, and other assessment as per Schedule of Assessments ([Section 7.1](#)).

If the subject stops treatment (eg, due to an Adverse Event), ensure there is a plan for continued follow-up until the subject completes all protocol-required visits or withdraws from the study. During this interval, there may be a reduction in the frequency and type of assessments required, preserving the end points of the study where possible.

7.2.3 End of Study Visit/ Early Termination

The following procedures will be completed at the EOS visits as per Schedule of Assessments ([Section 7.1](#)):

- Complete Physical Examination
- ECG
- Vitals
- Clinical Assessments of Disease Activity
 - Tender/Swollen 28-Joint Count
 - Patient Global Assessment of Disease Activity
 - Physician Global Assessment of Disease Activity
 - HAQ-DI
- Drug and Alcohol Screen
- Laboratory Assessments (chemistry, hematology, and urinalysis)
- Pregnancy Testing (females only)
- C-Reactive Protein (CRP)
- Erythrocyte Sedimentation Rate (ESR)
- Biomarker and Immunological Assessments

The EOS for each cohort may be prolonged pending treatment emergent data.

If a subject terminates enrollment in the study early, women of child-bearing potential and men should be counseled to continue contraception for a period of time equivalent to 5 half-lives after receiving the terminal dose of IP.

7.3 General Study Assessments

7.3.1 Physical Exam

The investigator or designated site physician will perform a complete physical examination (excluding breast, rectal examination, and genital examination) including height (screening only) and weight measurements at screening and at timepoints specified in the Schedule of Assessments ([Section 7.1](#)).

The individual performing the physical examination at screening and on admission to the research facility will characterize their findings as either normal or abnormal. Abnormal findings found during screening or baseline will be reported on the medical history page of the eCRF. Once enrolled, any change from the baseline physical examination, which represents deterioration, will be documented on the adverse event page of the eCRF.

Height measurement (in cm and without shoes) and weight measurement (in kg and without shoes) will be obtained.

7.3.2 Vital Signs

The following measurements must be performed: systolic/diastolic blood pressure (BP), heart rate (HR), respiration rate, and temperature. Subject must be in a supine position in a rested and calm state for at least 5 minutes before BP assessments are conducted. If the subject is unable to be in the supine position, the subject should be in most recumbent position as possible. The position selected for a subject should be the same that is used throughout the study and documented on the vital sign eCRF.

Record all measurements on the vital signs eCRF. The temperature location selected for a subject should be the same that is used throughout the study and documented on the vital signs/temperature log eCRF.

Vital signs will be recorded by the investigator or designated site physician at screening and time points specified in the Schedule of Assessments ([Section 7.1](#)).

Abnormal measurements may be repeated at the discretion of the investigator and must be reported on the corresponding eCRF page. When vital signs and blood sample collection occur at the same time, vital signs should be performed before blood samples are drawn.

7.3.3 Electrocardiogram

Electrocardiograms will be performed by the PI or designee at the time points listed in the Schedule of Assessments ([Section 7.1](#)). Subjects must be supine for at least 5 minutes before each ECG is obtained. All ECGs must be done in triplicate (approximately 1 minute apart). ECGs should be obtained before blood samples are drawn. On Day 1 pre-dose, triplicate ECGs will be performed on 3 occasions separated by at least 30 minutes for a total of 9 ECGs on Day 1 (ie, completed at 90, 60, and 30 minutes prior to dosing) and will serve as baseline. The original ECG tracings [reporting parameters including ventricular rate, PR, QRS, QT, QTc (corrected QT) and QTcF (QT interval corrected for heart rate using Fridericia's formula) intervals] will be

retained in the source documents. The principal investigator or designated physician will review all ECGs. Digital copies of ECGs for all subjects will be made available to Amgen.

7.3.4 Adverse Events and Concomitant Medications

Subjects will be assessed for adverse events ([Section 9](#)) and concomitant medication(s) throughout the study at each study visit. Any adverse event or concomitant medication use reported throughout the study will be recorded in the source documentation and on the subject's corresponding eCRF.

7.4 Disease Assessments

7.4.1 DAS28-CRP Score

The Disease Activity Score using 28 joints (DAS28) will be completed as listed in the Schedule of Assessments ([Section 7.1](#)) and will be calculated by Amgen. The DAS 28 is a composite score that is based on 28-joint count (using tender and swollen joints), CRP, and Patient Global Assessment of Disease Activity.

Calculation of the DAS28-CRP score is as follows:

$$\text{DAS 28-CRP} = 0.56 \sqrt{\text{TJC28}} + 0.28 \sqrt{\text{SJC28}} + 0.36 \times \ln(\text{CRP} + 1) + 0.014 \times \text{Patient Global Assessment of Disease Activity} + 0.96$$

where TJC28 and SJC28 represent the tender and swollen 28 joint counts, respectively.

The 28 joint counts includes 10 proximal interphalangeal, 10 metacarpophalangeal, 2 wrist, 2 elbow, 2 shoulder, and 2 knee joints. CRP will be analyzed at a central laboratory as noted in [Section 7.10.1](#). Only screening CRP values will be released to sites for determination of eligibility. These values will be entered into a CRF associated with screening where the DAS28-CRP score will be derived.

Joint Assessments

All joint assessments will be performed by an experienced independent and blinded joint evaluator. The evaluator cannot be the treating physician and cannot interact with the subject throughout the study beyond the assessment of joints nor for any other subject assessments or laboratory measures of efficacy. The evaluator should not discuss the subject's clinical status nor should the evaluator have access to subject medical records or eCRFs including prior joint assessments. The investigator physician may perform the screening assessment, however, if so, the independent evaluator will be required to confirm eligibility with the baseline visit joint assessment.

Unevaluable Joints

Joints that are or have been replaced are considered unevaluable.

Tender Joint Count (TJC) Assessments

A total of 28 joints will be scored for presence or absence of tenderness.

Swollen Joint Count (SJC) Assessments

A total of 28 joints will be scored for presence or absence of swelling.

7.4.2 Physician Global Assessment of Disease Activity

The treating physician will rate the patient's arthritis by marking an "X" between 0 and 100 on the 100 mm horizontal line (visual analog scale). The physician evaluator and subject must complete the global assessments independently from each other. The physician assessing the subject's global disease activity will have access to the joint assessments.

7.4.3 Patient Global Assessment of Disease Activity

The patient will rate the patient's arthritis status by marking an "X" between 0 and 100 on the 100 mm horizontal line (visual analog scale). The physician evaluator and subject must complete the global assessments independently from each other. This questionnaire (along with the HAQ-DI) will be given to subjects to complete before any other study procedures are performed on the subjects on the day of assessments.

See [Appendix D](#) for sample of the Patient Global Assessment of Disease Activity questionnaire.

7.4.4 Patient's Assessment of Physical Function: Disability Index of the Health Assessment Question (HAQ-DI)

The subject's self assessment of physical function or disability will be assessed by the HAQ-DI. The HAQ-DI was developed as a comprehensive measure of outcome in subjects with a wide variety of rheumatic diseases, including RA. Its focus is on self-reported subject-oriented outcome measures. HAQ-DI (along with the Patient's Assessment of Disease Activity) will be given to subjects to complete before any other procedures are performed on the subjects on the day of the assessment. This questionnaire asks about the degree of difficulty a person has in accomplishing task in 8 functional areas (dressing, arising, eating, walking, hygiene, reaching, gripping and errands and chores). Responses in each functional area are scored from 0 indicating no difficulty, to 3, indicating inability to perform a task in that area. The subject should be instructed to read each question carefully and mark an "X" for the option that best

represents their answer. The study staff should not clarify any of the questions for the subject, if asked. See [Appendix D](#) for sample of the Disability Index of the Health Assessment Question (HAQ-DI).

7.5 Blood and Urine Assessments

7.5.1 Blood Volume

Subjects enrolled in this study will provide whole blood for safety, PK, PD and biomarker assessments during their participation in this study. Estimated volume of blood collection is indicated in [Table 6](#). The total blood volume per subject is approximately 900 mL.

7.5.2 Drug and Alcohol Screen

A urine screen for drugs of abuse (cannabis, cocaine, amphetamines, benzodiazepines, opiates, barbiturates) and urine (or breath) ethanol screens will be performed at time points specified in the Schedule of Assessments ([Section 7.1](#)). Subjects with a positive test result will be excluded from the study, unless medication is prescribed by a physician and approved by the investigator and Amgen medical monitor.

7.5.3 HIV Antibodies, Hepatitis B Surface Antigen, and Hepatitis C Antibodies

Hepatitis B surface antigen (HepBsAg), hepatitis B core antibody (HBcAb), hepatitis C HepCAb, and HIV status will be assessed. If the results show a negative HepBsAg and positive for HBcAb, a positive hepatitis B virus DNA by polymerase chain reaction (PCR) is necessary.

If the results show a positive HepCAb: hepatitis C virus ribonucleic acid (RNA) by PCR is necessary. The test must be confirmed negative at screening for the subject to be eligible for this study.

7.5.4 Laboratory Safety Tests

Table 6. List of Analytes

Local Laboratory Chemistry (approx. 8 mL)	Local Laboratory Hematology (approx. 5 mL)	Local Laboratory Urinalysis	Other (approx. 3-8 mL)
Alanine aminotransferase (ALT)	Erythrocyte Sedimentation Rate (ESR)	Specific gravity	Central Laboratory:
Albumin	Hemoglobin	pH	CRP
Alkaline phosphatase	Hematocrit	Blood	Antibodies
Aspartate aminotransferase (AST)	Mean corpuscular volume (MCV)	Protein	Hepatitis B surface antigen
Bicarbonate (HCO ₃) or Carbon Dioxide	Platelets	Glucose	Hepatitis C antibody
Bilirubin	Reticulocyte count	Bilirubin	HIV
• Total	White blood cell count	Urobilinogen	
• Direct	White blood cell Differential (in absolute cell numbers)	Microscopic exam (performed at the discretion of the Investigator or designee):	Local Laboratory: QuantiFERON-TB Gold**
Blood urea nitrogen or Urea	• Total neutrophils	• White blood cells	FSH/Estradiol**
Calcium	• Eosinophils	• Red blood cells	Serum beta-HCG and/or Urine Pregnancy***
Chloride	• Basophils	• Epithelial cells	
Cholesterol	• Lymphocytes	• Bacteria	
Creatinine	• Monocytes	• Casts	
Creatine phosphokinase (CPK)	• RBC count	• Crystals	
Gamma-glutamyl transferase (GGT)	• MCV		
Glucose	• MCH		
High-density lipoprotein	• MCHC		
Magnesium			
Phosphorus			
Potassium			
Sodium			
Total protein			
Triglycerides			
Uric acid			
Hemoglobin A1C**			

*All subjects must fast from all food and drink (except water) for at least 8 hours prior to any clinical lab evaluation

**Screening only

***Per schedule of assessments ([Section 7.1](#))

7.5.5 Serum and Urine Pregnancy Test

A serum pregnancy test will be conducted for all women at Screening. A urine pregnancy test will be performed on Screening and days -1, 85, 239 (EOS). The results of these tests will be retained in the source document. In the event a female subject

becomes pregnant while on study, a pregnancy notification worksheet ([Appendix C](#)) must be completed and forwarded to the Sponsor.

7.5.6 FSH and Estradiol

Additional serum will be collected for an FSH and estradiol for females who are at least 1 year postmenopausal at screening unless 60 years or older. Results must be consistent with postmenopausal status per local laboratory ranges to be eligible for this study. Postmenopausal status will be recorded on the medical history eCRF.

7.5.7 QuantiFERON-TB Gold

A QuantiFERON®-TB Gold test will be completed during screening for all subjects, and results must be negative to be eligible for this study.

7.6 Urinalysis

All laboratory tests must be reviewed and signed by the principal investigator or qualified designee. Additional safety laboratory assessments may be performed for subject safety. Urine samples will be collected at time points specified in the Schedule of Assessments ([Section 7.1](#)).

The tests listed below will be conducted and analyzed by standard laboratory procedures.

7.7 Immunophenotyping and Receptor Occupancy Assay

Whole blood samples for Immunophenotyping and Receptor Occupancy (IPRO) will be collected at timepoints indicated in the Schedule of Assessments ([Section 7.1](#)). The Immunophenotyping assay will measure T, B, and NK cells and B cell subsets. The Receptor Occupancy assay will measure ICOSL receptor occupancy. Instructions for collecting, processing, and shipping the IPRO samples will be provided in a separate document from Amgen. Immunophenotyping blood collection instructions for collecting, processing, and shipping samples are provided in a separate document from Amgen.

7.8 Serum Immunoglobulin Levels

Blood samples will be collected for measurement of serum Immunoglobulins (IgM, IgG and IgA) at time points indicated in the Schedule of Assessments ([Section 7.1](#)). Instructions for collecting, processing, and shipping samples will be provided in a separate document from Amgen.

7.9 BAFF Serum Levels

Blood samples will be collected for determination of serum BAFF levels at time points indicated in the Schedule of Assessments ([Section 7.1](#)). Instructions for collecting, processing, and shipping samples will be provided in a separate document from Amgen.

7.10 Rheumatoid Arthritis Biomarkers

7.10.1 C-Reactive Protein

Blood samples for CRP will be collected at timepoints according to the Schedule of Assessments ([Section 7.1](#)). The central laboratory will provide CRP results to sites at screening for eligibility assessment.

7.10.2 Rheumatoid Factor

A blood sample for determination of RF will be collected and analyzed at a central laboratory. Total RF and isotypes will be assessed.

7.10.3 Anti-citrullinated Peptide Antibodies

A blood sample will be collected to measure antibodies against citrullinated peptides at timepoints specified in the Schedule of Assessments ([Section 7.1](#)) and analyzed at a central laboratory.

7.11 Exploratory Autoimmune Antibodies

A blood sample will be collected for exploratory autoimmune antibody analysis at timepoints specified in the Schedule of Assessments ([Section 7.1](#)).

7.12 Blood Collection for Pharmacokinetic Analysis

A blood sample will be collected for pharmacokinetic analysis at timepoints specified in the Schedule of Assessments ([Section 7.1](#)).

7.13 Antibody Testing Procedures

Blood samples will be collected as outlined in the Schedule of Assessments ([Section 7.1](#)) for the measurement of anti-AMG 570 binding antibodies. Samples testing positive for binding antibodies may be further characterized for quantity/titer, isotype, affinity, or in vitro neutralizing activity. Additional blood samples may be obtained to evaluate any anti-AMG 570 antibody mediated impact on safety, PK and/or PD, and efficacy during the study. Subjects who test positive for binding antibodies and have clinical sequelae that are considered potentially related to an anti-AMG 570 antibody response may also be asked to return for additional follow-up testing.

7.14 Biomarker Development

Biomarkers are objectively measured and evaluated indicators of normal biologic processes, pathogenic processes, or pharmacologic responses to a therapeutic intervention. Biomarker development can be useful in developing markers to identify disease subtypes, guide therapy, and/or predict disease severity.

Amgen may attempt to develop test(s) designed to identify subjects most likely to respond positively or negatively to investigational product(s) (eg, Amgen or non-Amgen investigational product or protocol required therapies to investigate and further understand RA.

Blood Samples

Blood samples are to be collected for biomarker development as specified in Schedule of Assessments ([Section 7.1](#)).

7.15 Pharmacogenetic Studies

If the subject consents to the optional pharmacogenetic portion of this study, DNA analyses may be performed. These optional pharmacogenetic analyses focus on inherited genetic variations to evaluate their possible correlation to the disease and/or responsiveness to the therapies used in this study. The goals of the optional studies include the use of genetic markers to help in the investigation of RA and/or to identify subjects who may have positive or negative response to investigational product or protocol required therapies. No additional samples are collected for this part of the study. For subjects who consent to this/these analysis/analyses, DNA may be extracted.

7.16 Sample Storage and Destruction

Any blood (eg, biomarker, PK) sample collected according to the Schedule of Assessments ([Section 7.1](#)) can be analyzed for any of the tests outlined in the protocol and for any tests necessary to minimize risks to study subjects. This includes testing to ensure analytical methods produce reliable and valid data throughout the course of the study. This can also include, but is not limited to, investigation of unexpected results, incurred sample reanalysis, and analyses for method transfer and comparability.

All samples and associated results will be coded prior to being shipped from the site for analysis or storage. Samples will be tracked using a unique identifier that is assigned to the samples for the study. Results are stored in a secure database to ensure confidentiality.

If informed consent is provided by the subject, Amgen can do additional testing on remaining samples (ie, residual and back-up) to investigate and better understand RA, the dose response and/or prediction of response to AMG 570, as required, and characterize aspects of the molecule (eg, mechanism of action/target, metabolites). Results from this analysis are to be documented and maintained, but are not necessarily reported as part of this study. Samples can be retained for up to 20 years.

Since the evaluations are not expected to benefit the subject directly or to alter the treatment course, the results of pharmacogenetic, biomarker development, or other exploratory studies are not placed in the subject's medical record and are not to be made available to the subject, members of the family, the personal physician, or other third parties, except as specified in the informed consent.

The subject retains the right to request that the sample material be destroyed by contacting the investigator. Following the request from the subject, the Investigator is to provide the sponsor with the required study and subject number so that any remaining sample types (eg, blood, tumor) samples and any other components from the cells can be located and destroyed. Samples will be destroyed once all protocol-defined procedures are completed. However, information collected from samples prior to the request for destruction, will be retained by Amgen.

The sponsor is the exclusive owner of any data, discoveries, or derivative materials from the sample materials and is responsible for the destruction of the sample(s) at the request of the subject through the investigator, at the end of the storage period, or as appropriate (eg, the scientific rationale for experimentation with a certain sample type no longer justifies keeping the sample). If a commercial product is developed from this research project, the sponsor owns the commercial product. The subject has no commercial rights to such product and has no commercial rights to the data, information, discoveries, or derivative materials gained or produced from the sample.

See [Section 11.3](#) for subject confidentiality.

8. WITHDRAWAL FROM TREATMENT, PROCEDURES, AND STUDY

8.1 Subjects' Decision to Withdraw

Subjects have the right to withdraw from the study at any time and for any reason without prejudice to their future medical care by the physician or at the institution.

Subjects (or a legally acceptable representative) can decline to continue receiving investigational product and/or other protocol-required therapies or procedures at any

time during the study but continue participation in the study. If this occurs, the investigator is to discuss with the subject the appropriate processes for discontinuation from investigational product, device or other protocol-required therapies and must discuss with the subject the options for continuation of the Schedule of Assessments ([Section 7.1](#)) including different options of follow-up (eg, in person, by phone/mail, through family/friends, in correspondence/communication with other treating physicians, from the review of medical records) and collection of data, including endpoints, adverse events, disease related events. Subjects who have discontinued investigational product and/or protocol required therapies or procedures should not be automatically removed from the study. Whenever safe and feasible it is imperative that subjects remain on-study to ensure safety surveillance and/or collection of outcome data. The investigator must document the level of follow-up that is agreed to by the subject.

Withdrawal of consent for a study means that the subject does not wish to receive further protocol-required therapies or procedures, and the subject does not wish to or is unable to continue further study participation. Subject data up to withdrawal of consent will be included in the analysis of the study, and where permitted, publically available data can be included after withdrawal of consent. The investigator is to discuss with the subject appropriate procedures for withdrawal from the study.

8.2 Investigator or Sponsor Decision to Withdraw or Terminate Subjects' Participation Prior to Study Completion

The investigator and/or sponsor can decide to withdraw a subject(s) from investigational product, medical device(s), and/or other protocol required therapies, protocol procedures, or the study as a whole at any time prior to study completion.

Subjects may be eligible for continued treatment with Amgen investigational product(s) and/or other protocol required therapies by a separate protocol or as provided for by the local country's regulatory mechanism, based on parameters consistent with [Section 12.1](#).

8.3 Reasons for Removal From Treatment and/or Study

8.3.1 Reasons for Removal From Treatment

Reasons for removal from protocol-required investigational product(s) or procedural assessments include any of the following:

- subject request
- safety concern (eg, due to an adverse event, ineligibility determined, protocol deviation, non-compliance, requirement for alternative therapy, protocol-specified criteria (list criteria), pregnancy)
- death
- lost to follow-up
- decision by sponsor (other than subject request, safety concern, lost to follow-up)

8.3.2 Reasons for Removal From Study

Reasons for removal of a subject from the study are:

- decision by sponsor
- withdrawal of consent from study
- death
- lost to follow-up

9. SAFETY DATA COLLECTION, RECORDING, AND REPORTING

9.1 Definition of Safety Events

9.1.1 Adverse Events

An adverse event is defined as any untoward medical occurrence in a clinical trial subject. The event does not necessarily have a causal relationship with study treatment. The investigator is responsible for ensuring that any adverse events observed by the investigator or reported by the subject are recorded in the subject's medical record.

The definition of adverse events includes worsening of a pre-existing medical condition. Worsening indicates that the pre-existing medical condition or underlying disease (eg, diabetes, migraine headaches, gout) has increased in severity, frequency, and/or duration more than would be expected and/or has an association with a significantly worse outcome than expected. A pre-existing condition that has not worsened more than anticipated (ie, more than usual fluctuation of disease) during the study, or involves an intervention such as elective cosmetic surgery or a medical procedure while on study, is not considered an adverse event.

If the severity of an adverse event changes from the date of onset to the date of resolution, record as a single event with the worst severity on the Event CRF. The

investigator's clinical judgment is used to determine whether a subject is to be removed from treatment due to an adverse event. In the event a subject, or subject's legally acceptable representative requests to withdraw from protocol-required therapies or the study due to an adverse event, refer to [Section 8.1](#) for additional instructions on the procedures recommended for safe withdrawal from protocol-required therapies or the study.

9.1.2 Serious Adverse Events

A serious adverse event is defined as an adverse event that meets at least 1 of the following serious criteria (unless it meets the definition of a Disease-Related Event as defined in [Section 9.1.1](#)):

- fatal
- life threatening (places the subject at immediate risk of death)
- requires in patient hospitalization or prolongation of existing hospitalization
- results in persistent or significant disability/incapacity
- congenital anomaly/birth defect
- other medically important serious event

An adverse event would meet the criterion of "requires hospitalization", if the event necessitated an admission to a health care facility (eg, overnight stay).

If an investigator considers an event to be clinically important, but it does not meet any of the serious criteria, the event could be classified as a serious adverse event under the criterion of "other medically important serious event". Examples of such events could include allergic bronchospasm, convulsions, blood dyscrasias, drug induced liver injury (DILI) (see [Appendix A](#) for DILI reporting criteria), or events that necessitate an emergency room visit, outpatient surgery, or urgent intervention.

9.2 Safety Event Reporting Procedures

9.2.1 Adverse Events

9.2.1.1 Reporting Procedures for Adverse Events That do not Meet Serious Criteria

The investigator is responsible for ensuring that all adverse events observed by the investigator or reported by the subject that occur after enrollment through the end of study are reported using the Event CRF.

The investigator must assign the following adverse event attributes:

- Adverse event diagnosis or syndrome(s), if known (if not known, signs or symptoms),
- Dates of onset and resolution (if resolved),
- Severity (using CTCAE scale)
- Assessment of relatedness to investigational product, and
- Action taken.

The adverse event grading scale used will be the Common Terminology Criteria for Adverse Events (CTCAE) Version 4. The grading scale used in this study is described in [Appendix A](#). The investigator must assess whether the adverse event is possibly related to the investigational product. This relationship is indicated by a “yes” or “no” response to the question: Is there a reasonable possibility that the event may have been caused by the investigational product?

The investigator must assess whether the adverse event is possibly related to any study mandated activity and/or procedure. This relationship is indicated by a “yes” or “no” response to the question: “Is there a reasonable possibility that the event may have been caused by a study activity and/or procedure”?

The investigator is responsible for reviewing laboratory test results and determining whether an abnormal value in an individual study subject represents a clinically significant change from the subject’s baseline values. In general, abnormal laboratory findings without clinical significance (based on the Investigator’s judgment) are not to be recorded as adverse events. However, laboratory value changes that require treatment or adjustment in current therapy are considered adverse events. Where applicable, clinical sequelae (not the laboratory abnormality) are to be recorded as the adverse event.

The investigator is expected to follow reported adverse events until stabilization or reversibility.

9.2.1.2 Reporting Procedures for Serious Adverse Events

The investigator is responsible for ensuring that all serious adverse events observed by the investigator or reported by the subject that occur after signing of the informed consent through 30 days after the last administered dose of the study drug. All serious adverse events must be submitted to Amgen within 24 hours following the investigator’s knowledge of the event via the Event CRF. All Grade 4 (life-threatening) events should be reported as Serious Adverse Events.

If the electronic data capture (EDC) system is unavailable to the site staff to report the serious adverse event, the information is to be reported to Amgen via an electronic Serious Adverse Event (eSAE) Contingency Report Form within 24 hours of the investigator's knowledge of the event. See [Appendix B](#) for a sample of the Serious Adverse Event Worksheet/electronic Serious Adverse Event Contingency Report Form. For EDC studies where the first notification of a Serious Adverse Event is reported to Amgen via the Electronic Serious Adverse Event Contingency Report Form, the data must be entered into the EDC system when the system is again available.

The investigator must assess whether the serious adverse event is possibly related to the investigational product. This relationship is indicated by a "yes" or "no" response to the question: Is there a reasonable possibility that the event may have been caused by the investigational product? Relatedness means that there are facts or reasons to support a relationship between investigational product and the event.

The investigator must assess whether the serious adverse event is possibly related to any study mandated activity and/or procedure (including any screening procedure(s)). This relationship is indicated by a "yes" or "no" response to the question: "Is there a reasonable possibility that the event may have been caused by a study activity and/or procedure"?

The investigator is expected to follow reported serious adverse events until stabilization or reversibility.

New information relating to a previously reported serious adverse event must be submitted to Amgen. All new information for serious adverse events must be sent to Amgen within 24 hours following knowledge of the new information. If specifically requested, the investigator may need to provide additional follow-up information, such as discharge summaries, medical records, or extracts from the medical records. Information provided about the serious adverse event must be consistent with that recorded on the Event CRF.

If a subject is permanently withdrawn from protocol-required therapies because of a serious adverse event, this information must be submitted to Amgen.

To comply with worldwide reporting regulations for serious adverse events, the treatment assignment of subjects who develop serious, unexpected, and related adverse events may be unblinded by Amgen before submission to regulatory authorities.

Approved

Amgen will report serious adverse events and/or suspected unexpected serious adverse reactions as required to regulatory authorities, investigators/institutions, and IRBs/IECs in compliance with all reporting requirements according to local regulations and good clinical practice.

The investigator is to notify the appropriate IRB/IEC of serious adverse events occurring at the site and other adverse event reports received from Amgen, in accordance with local regulatory requirements and procedures.

9.2.1.3 Reporting Serious Adverse Events After the Protocol-required Reporting Period

There is no requirement to monitor study subjects for serious adverse events following the protocol-required reporting period or after end of study. However, these serious adverse events can be reported to Amgen. In some countries (eg, European Union [EU] member states), investigators are required to report serious adverse events that they become aware of after end of study. If serious adverse events are reported, the investigator is to report them to Amgen within 24 hours following the investigator's knowledge of the event.

Serious adverse events reported outside of the protocol-required reporting period will be captured within the safety database as clinical trial cases for the purposes of expedited reporting.

9.3 Pregnancy and Lactation Reporting

If a female subject becomes pregnant, or a male subject fathers a child, while the subject is taking investigational product (AMG 570 or placebo), report the pregnancy to Amgen Global Patient Safety as specified below.

In addition to reporting any pregnancies occurring during the study, investigators should report pregnancies that occur through 20 weeks after the last dose of AMG 570.

The pregnancy should be reported to Amgen Global Patient Safety within 24 hours of the investigator's knowledge of the pregnancy. Report a pregnancy on the Pregnancy Notification Worksheet ([Appendix C](#)). Amgen Global Patient Safety will follow-up with the investigator regarding additional information that may be requested.

If a female subject becomes pregnant during the study, the investigator should attempt to obtain information regarding the birth outcome and health of the infant.

If the outcome of the pregnancy meets a criterion for immediate classification as a Serious Adverse Event (eg, female subject experiences a spontaneous abortion,

stillbirth, or neonatal death or there is a fetal or neonatal congenital anomaly) the investigator will report the event as a Serious Adverse Event.

If a female breastfeeds while taking protocol-required therapies report the lactation case to Amgen as specified below.

In addition to reporting a lactation case during the study, investigators should report lactation cases that occur through 20 weeks after the last dose of protocol-required therapies.

Any lactation case should be reported to Amgen Global Patient Safety within 24 hours of the Investigator's knowledge of event. Report a lactation case on the Lactation Notification Worksheet ([Appendix C](#)). Amgen Global Patient Safety will follow-up with the investigator regarding additional information that may be requested.

If a male subject's female partner becomes pregnant, the investigator should discuss obtaining information regarding the birth outcome and health of the infant from the pregnant partner.

10. STATISTICAL CONSIDERATIONS

10.1 Study Endpoints, Analysis Sets, and Covariates

10.1.1 Study Endpoints

Primary Endpoint:

Subject incidence of treatment-emergent adverse events (including clinically significant changes in physical examinations), vital signs, laboratory safety tests, and electrocardiograms (ECGs).

Secondary Endpoint(s):

- AMG 570 Serum PK profiles after multiple dose administrations. PK parameters will include, but are not limited to, maximum observed concentration [C_{max}] and area under the concentration-time curve [AUC] after the first and the last doses, and the accumulation ratio.

Exploratory Endpoint(s):

- Peripheral blood ICOSL receptor occupancy
- Baseline levels, post-dose levels and change from baseline in peripheral blood percentage and absolute counts of total T cells, B cells and NK cells
- Baseline levels, post-dose levels and change from baseline in peripheral blood percentage and absolute counts of naive and memory B cells (naïve = CD19+IgD+CD27-, memory = CD19+IgD-CD27+)
- Baseline levels, post-dose levels and change from baseline in circulating Plasma cells by gene signature

- Baseline levels, post-dose levels and change from baseline in circulating Tfh by flow cytometry and gene signature
- Baseline levels, post-dose levels and change from baseline in serum Immunoglobulin M, Immunoglobulin G and Immunoglobulin A
- Baseline levels, post-dose levels and change from baseline in serum BAFF
- Baseline levels, post-dose levels and change from baseline in serum CRP, ESR, IgG+ Rheumatoid Factor, IgM+ Rheumatoid Factor and anti-citrullinated protein antibody
- Changes from baseline over time in the disease activity score based on 28 joints (DAS 28-CRP)
- Changes from baseline over time in Patient Global Assessment of Disease activity
- Changes from baseline over time in Physician Global Assessment of Disease activity
- Anti-AMG 570 binding antibodies

10.1.2 Analysis Sets

The safety analysis set will include all randomized subjects who received at least one dose of AMG 570/placebo.

The PK concentration analysis set will contain all randomized subjects who received AMG 570 and have at least one quantifiable PK sample collected. The PK parameter analysis set will contain all randomized subjects who received AMG 570 and for whom PK parameters can be adequately estimated.

The PD analysis set will contain all randomized subjects who received at least 1 dose of investigational product (AMG 570 or Placebo) and for whom at least 1 PD parameter has baseline and at least 1 post baseline measurement available.

10.1.3 Covariates and Subgroups

No subgroup analysis is planned.

10.1.4 Handling of Missing and Incomplete Data

Statistical analysis plan will detail the handling of missing and incomplete data.

10.2 Sample Size Considerations

Approximately 32 RA patients will be enrolled into 4 cohorts with 6 active and 2 placebo in each cohort.

This sample size is based on practical considerations and is typical for this type of study. For safety considerations, for a cohort, with 6 subjects receiving AMG 570, there is an 82% chance of at least one subject experiencing an adverse event with a true incidence

rate of 25% and a 74% chance of at least one subject experiencing an adverse event with 20% true incidence rate. With a total of 18 subjects expected to receive AMG 570 across all 3 cohorts, there is a 17% chance of at least 1 subject experiencing an adverse event with a true incidence rate of 1% and the chance of at least 1 subject experiencing an adverse event increases to 60% and 85% with a true incidence rate of 5% and 10%, respectively.

10.3 Access to Individual Subject Treatment Assignments by Amgen or Designees

Amgen staff and their designees involved in the study will not be blinded, but will be given access to the unblinding or potentially unblinding information only when there is a need to use the information for analysis discussion and internal decision-making, in particular, when concerning safety issues. Access to treatment assignments and other restricted data are described in Amgen standard documents. Unblinded individuals will ensure the keeping of the blind. Unblinding and potentially unblinding information should not be distributed to the investigators or subjects prior to a study cohort being formally unblinded.

10.4 Planned Analyses

10.4.1 Data Monitoring Committee (DMC), Data Review Team (DRT) or Dose Level Review Team (DLRT)

A Dose Level Review Team (DLRT) will be used to oversee progress of the study and make recommendations relating to early closure/extension or alteration of the study based on ongoing monitoring of the study data. The DLRT members will consist of the principal investigator or designee, Amgen Medical Monitor, Amgen Global Safety Officer or designee, Amgen Clinical Research Study Manager or designee, and biostatistics representative or designee. Additional members may be added as needed (eg, PK scientist). The key objectives of the DLRT are to review data, monitor safety, and make dose change decisions.

10.4.2 Primary Analysis

The primary analysis will occur after all subjects have completed the study.

10.5 Planned Methods of Analysis

10.5.1 General Considerations

Descriptive statistics will be provided for selected demographics, safety, immunogenicity, PK, PD, and biomarker data. Descriptive statistics on continuous measurements will include means, medians, standard deviations, and ranges, while categorical data will be

summarized using frequency counts and percentages. Data will be summarized by treatment and by scheduled timepoint unless specified otherwise.

10.5.2 Primary Endpoint(s)

Safety analysis set will be used for analyses included in this section.

10.5.2.1 Adverse Event

Subject incidence of all treatment-emergent adverse events (including clinically significant changes in physical examinations) will be tabulated by system organ class and preferred term. Adverse events will be coded using the Medical Dictionary for Regulatory Activities. Tables of treatment related adverse events, fatal adverse events, serious adverse events, and adverse events leading to withdrawal from investigational product or other protocol-required therapies or from study will also be provided. Subject incidence of treatment emergent events and treatment related treatment emergent events will further be summarized by worst severity grade.

10.5.2.2 Clinical Laboratory Tests

The analyses of safety laboratory endpoints may include summary statistics over time and/or changes from baseline. Shifts in grades of selected analytes between the baseline and the worst on-study value may be tabulated.

10.5.2.3 Vital Signs

Summary statistics over time and/or changes from baseline over time in vital signs will be provided.

10.5.2.4 Electrocardiograms

Descriptive summary statistics of ECG values over time and/or changes from baseline over time may be provided for each ECG parameter. Subjects' maximum change from baseline in QTcF will be categorized and the number and percentage of subjects in each category will be summarized. Subjects' maximum post baseline values will also be categorized and the number and percentage of subjects in each category will be summarized.

10.5.3 Secondary Endpoint(s)

10.5.3.1 PK Analysis

Serum AMG 570 concentrations will be determined using a validated assay. Individual serum concentration-time plots for AMG 570 will be presented for each subject as well as mean concentration-time plots for each treatment.

Pharmacokinetic parameters (eg, maximum observed concentration [C_{max}], time at C_{max} [t_{max}], and area under the concentration-time curve [AUC]) will be calculated using non-compartmental methods. Actual dosing and sampling times will be used for calculation of individual PK parameters. Summary statistics will be generated for each PK parameter.

Additional analyses using compartmental methods or population PK analysis may be performed.

10.5.4 Exploratory Endpoints

All exploratory end points will be summarized using descriptive statistics. The change from baseline and/or fold change from baseline also may be summarized.

The incidence and percentage of subjects who develop anti-AMG 570 antibodies (binding and if positive, neutralizing) at any time will be tabulated.

11. REGULATORY OBLIGATIONS

11.1 Informed Consent

An initial sample informed consent form is provided for the investigator to prepare the informed consent document to be used at his or her site. Updates to the template are to be communicated formally in writing from the Amgen Study Manager to the investigator. The written informed consent form is to be prepared in the language(s) of the potential subject population.

Before a subject's participation in the clinical study, the investigator is responsible for obtaining written informed consent from the subject after adequate explanation of the aims, methods, anticipated benefits, and potential hazards of the study and before any protocol specific screening procedures or any investigational product(s) is/ are administered.

The investigator is also responsible for asking the subject if the subject has a primary care physician and if the subject agrees to have his/her primary care physician informed of the subject's participation in the clinical study. If the subject agrees to such notification, the investigator is to inform the subject's primary care physician of the subject's participation in the clinical study. If the subject does not have a primary care physician and the investigator will be acting in that capacity, the investigator is to document such in the subject's medical record. The acquisition of informed consent and the subject's agreement or refusal of his/her notification of the primary care physician is to be documented in the subject's medical records, and the informed consent form is to

be signed and personally dated by the subject and by the person who conducted the informed consent discussion. The original signed informed consent form is to be retained in accordance with institutional policy, and a copy of the signed consent form is to be provided to the subject.

If a potential subject is illiterate or visually impaired and does not have a legally acceptable representative, the investigator must provide an impartial witness to read the informed consent form to the subject and must allow for questions. Thereafter, both the subject and the witness must sign the informed consent form to attest that informed consent was freely given and understood.

11.2 Institutional Review Board/Independent Ethics Committee

A copy of the protocol, proposed informed consent form, other written subject information, and any proposed advertising material must be submitted to the IRB/IEC for written approval. A copy of the written approval of the protocol and informed consent form must be received by Amgen before recruitment of subjects into the study and shipment of Amgen investigational product.

The investigator must submit and, where necessary, obtain approval from the IRB/IEC for all subsequent protocol amendments and changes to the informed consent document. The investigator is to notify the IRB/IEC of deviations from the protocol or serious adverse events occurring at the site and other adverse event reports received from Amgen, in accordance with local procedures.

The investigator is responsible for obtaining annual IRB/IEC approval/renewal throughout the duration of the study. Copies of the investigator's reports and the IRB/IEC continuance of approval must be sent to Amgen.

11.3 Subject Confidentiality

The investigator must ensure that the subject's confidentiality is maintained:

- Subjects are to be identified by a unique subject identification number.
- Where permitted, date of birth is to be documented and formatted in accordance with local laws and regulations.
- On the demographics page, in addition to the unique subject identification number, include the age at the time of enrollment.

- For Serious Adverse Events reported to Amgen, subjects are to be identified by their unique subject identification number, initials (for faxed reports, in accordance with local laws and regulations), and date of birth (in accordance with local laws and regulations).
- Documents that are not submitted to Amgen (eg, signed informed consent forms) are to be kept in strict confidence by the investigator, except as described below.

In compliance with governmental/ICH GCP Guidelines, it is required that the investigator and institution permit authorized representatives of the company, of the regulatory agency(s), and the IRB/IEC direct access to review the subject's original medical records for verification of study related procedures and data. Direct access includes examining, analyzing, verifying, and reproducing any records and reports that are important to the evaluation of the study. The investigator is obligated to inform and obtain the consent of the subject to permit named such individuals to have access to his/her study related records, including personal information.

11.4 Investigator Signatory Obligations

Each clinical study report is to be signed by the investigator or, in the case of multi-center studies, the coordinating investigator.

The coordinating investigator, identified by Amgen, will be any or all of the following:

- a recognized expert in the therapeutic area
- an investigator who provided significant contributions to either the design or interpretation of the study
- an investigator contributing a high number of eligible subjects

12. ADMINISTRATIVE AND LEGAL OBLIGATIONS

12.1 Protocol Amendments and Study Termination

Amgen may amend the protocol at any time. After Amgen amends the protocol, the Investigator is to return the signed Investigator's Signature page confirming agreement to continue participation in the study according to the amendment. The IRB/IEC must be informed of all amendments and give approval. The investigator **must** send a copy of the approval letter from the IRB/IEC and amended protocol Investigator's Signature page to Amgen prior to implementation of the protocol amendment at their site.

Amgen reserves the right to terminate the study at any time. Both Amgen and the investigator reserve the right to terminate the Investigator's participation in the study according to the Clinical Trial Agreement. The investigator is to notify the IRB/IEC in writing of the study's completion or early termination and send a copy of the notification to Amgen.

Subjects may be eligible for continued treatment with Amgen investigational product by an extension protocol or as provided for by the local country's regulatory mechanism. However, Amgen reserves the unilateral right, at its sole discretion, to determine whether to supply Amgen investigational product(s), and by what mechanism, after termination of the study and before it is available commercially.

12.2 Study Documentation and Archive

The investigator is to maintain a list of appropriately qualified persons to whom he/she has delegated study duties. All persons authorized to make entries and/or corrections on CRFs will be included on the Amgen Delegation of Authority Form.

Source documents are original documents, data, and records from which the subject's CRF data are obtained. These include but are not limited to hospital records, clinical and office charts, laboratory and pharmacy records, diaries, microfiches, radiographs, and correspondence.

The Investigator and study staff are responsible for maintaining a comprehensive and centralized filing system of all study related (essential) documentation, suitable for inspection at any time by representatives from Amgen and/or applicable regulatory authorities.

Elements to include:

- Subject files containing completed CRF, informed consent forms, and subject identification list
- Study files containing the protocol with all amendments, Investigator's Brochure, copies of prestudy documentation, and all correspondence to and from the IRB/IEC and Amgen
- Investigational product-related correspondence including Proof of Receipts (POR), Investigational Product Accountability Record(s), Return of Investigational Product for Destruction Form(s), Final Investigational Product Reconciliation Statement, as applicable.
- Non-investigational product(s), and/or medical device(s) or combination product(s) documentation, as applicable.

In addition, all original source documents supporting entries in the CRFs must be maintained and be readily available.

Retention of study documents will be governed by the Clinical Trial Agreement.

12.3 Study Monitoring and Data Collection

The Amgen representative(s) and regulatory authority inspectors are responsible for contacting and visiting the investigator for the purpose of inspecting the facilities and,

upon request, inspecting the various records of the clinical study (eg, CRFs and other pertinent data) provided that subject confidentiality is respected.

The Clinical Monitor is responsible for verifying the CRFs at regular intervals throughout the study to verify adherence to the protocol; completeness, accuracy, and consistency of the data; and adherence to local regulations on the conduct of clinical research. The Clinical Monitor is to have access to subject medical records and other study related records needed to verify the entries on the CRFs.

The investigator agrees to cooperate with the clinical monitor to ensure that any problems detected in the course of these monitoring visits, including delays in completing CRFs, are resolved.

In accordance with ICH GCP and the sponsor's audit plans, this study may be selected for audit by representatives from Amgen's Global Compliance Auditing function (or designees). Inspection of site facilities (eg, pharmacy, protocol-required therapy storage areas, laboratories) and review of study related records will occur to evaluate the study conduct and compliance with the protocol, ICH GCP, and applicable regulatory requirements.

Data capture for this study is planned to be electronic:

- All source documentation supporting entries into the electronic CRFs must be maintained and readily available.
- Updates to electronic CRFs will be automatically documented through the software's "audit trail".
- To ensure the quality of clinical data across all subjects and sites, a clinical data management review is performed on subject data received at Amgen. During this review, subject data are checked for consistency, omissions, and any apparent discrepancies. In addition, the data are reviewed for adherence to the protocol and GCP. To resolve any questions arising from the clinical data management review process, data queries are created in the EDC system database for site resolution and subsequently closed by the EDC system or by an Amgen reviewer.
- The investigator signs only the Investigator Verification Form for this electronic data capture study. This signature indicates that the investigator inspected or reviewed the data on the CRF, the data queries, and agrees with the content.

Amgen (or designee) will perform Self-Evident Corrections (SEC) to obvious data errors in the clinical trial database. SECs will be documented in the CRF Standard Instructions and the CRF Specific Instructions, both of these will be available through the EDC system. Examples of obvious data errors that may be corrected by Amgen (or designee) include deletion of obvious duplicate data (ie, the same results sent twice with the same date with different visit, [eg, week 4 and early termination]) and updating a specific

response if the confirming datum is provided in the “other, specify” field (eg, for race, reason for ending study).

12.4 Investigator Responsibilities for Data Collection

The investigator is responsible for complying with the requirements for all assessments and data collection (including subjects not receiving protocol-required therapies) as stipulated in the protocol for each subject in the study. For subjects who withdraw prior to completion of all protocol-required visits and are unable or unwilling to continue the Schedule of Assessments ([Table 4](#)), the investigator can search publically available records [where permitted] to ascertain survival status. This ensures that the data set(s) produced as an outcome of the study is/are as comprehensive as possible.

12.5 Language

All written information and other material to be used by subjects and investigative staff must use vocabulary and language that are clearly understood.

12.6 Publication Policy

To coordinate dissemination of data from this study, Amgen may facilitate the formation of a publication committee consisting of several investigators and appropriate Amgen staff, the governance and responsibilities of which are set forth in a Publication Charter. The committee is expected to solicit input and assistance from other investigators and to collaborate with authors and Amgen staff as appropriate as defined in the Publication Charter. Membership on the committee (both for investigators and Amgen staff) does not guarantee authorship. The criteria described below are to be met for every publication.

Authorship of any publications resulting from this study will be determined on the basis of the International Committee of Medical Journal Editors (ICMJE) Recommendations for the Conduct of Reporting, Editing, and Publication of Scholarly Work in Medical Journals, which states:

- Authorship credit should be based on (1) substantial contributions to conception and design, acquisition of data, or analysis and interpretation of data; (2) drafting the article or revising it critically for important intellectual content; (3) final approval of the version to be published; and (4) agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. Authors should meet conditions 1, 2, and 3 and 4.
- When a large, multicenter group has conducted the work, the group should identify the individuals who accept direct responsibility for the manuscript. These individuals should fully meet the criteria for authorship defined above.
- Acquisition of funding, collection of data, or general supervision of the research group, alone, does not justify authorship.

- All persons designated as authors should qualify for authorship, and all those who qualify should be listed.
- Each author should have participated sufficiently in the work to take public responsibility for appropriate portions of the content.

All publications (eg, manuscripts, abstracts, oral/slide presentations, book chapters) based on this study must be submitted to Amgen for corporate review. The Clinical Trial Agreement among the institution, investigator, and Amgen will detail the procedures for, and timing of, Amgen's review of publications.

12.7 Compensation

Any arrangements for compensation to subjects for injury or illness that arises in the study are described in the Compensation for Injury section of the Informed Consent that is available as a separate document.

Approved

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14. APPENDICES

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Appendix A. Additional Safety Assessment Information

Adverse Event Grading Scale

The Common Terminology Criteria for Adverse Events (CTCAE) version 4.0 will be used and is available at the following location: <https://evs.nci.nih.gov/ftp1/CTCAE/About.html>.

Drug-induced Liver Injury Reporting & Additional Assessments

Reporting

To facilitate appropriate monitoring for signals of DILI, cases of concurrent AST or ALT and TBL and/or INR elevation according to the criteria specified in [Section 6.5](#) require the following:

- The event is to be reported to Amgen as a serious adverse event within 24 hours of discovery or notification of the event (ie, before additional etiologic investigations have been concluded)
- The appropriate CRF (eg, Adverse Event CRF) that captures information necessary to facilitate the evaluation of treatment-emergent liver abnormalities is to be completed and sent to the Amgen.

Other events of hepatotoxicity and potential DILI are to be reported as serious adverse events if they meet the criteria for a serious adverse event defined in [Section 9.2.1.2](#).

Additional Clinical Assessments and Observation

All subjects in whom investigational product(s) or protocol-required therapies is/are withheld (either permanently or conditionally) due to potential DILI as specified in [Section 6.5](#) or who experience AST or ALT elevations > 3 x ULN or 2-fold increase above baseline values for subjects with evaluated values before drug are to undergo a period of “close observation” until abnormalities return to normal or to the subject’s baseline levels. Assessments that are to be performed during this period include:

- Repeat AST, ALT, ALP, bilirubin (total and direct), and INR within 24 hours
- In cases of TBL > 2x ULN or INR > 1.5, retesting of liver tests, BIL (total and direct), and INR is to be performed every 24 hours until laboratory abnormalities improve
- Testing frequency of the above laboratory tests may decrease if the abnormalities stabilize or the investigational product(s) or protocol-required therapies has/have been discontinued AND the subject is asymptomatic.
- Initiate investigation of alternative causes for elevated AST or ALT and/or elevated TBL. The following are to be considered depending on the clinical situation:
 - Complete blood count (CBC) with differential to assess for eosinophilia
 - Serum total immunoglobulin IgG, Anti-nuclear antibody (ANA), Anti Smooth Muscle Antibody, and Liver Kidney Microsomal antibody 1 (LKM1) to assess for autoimmune hepatitis
 - Serum acetaminophen (paracetamol) levels

- A more detailed history of:
 - Prior and/or concurrent diseases or illness
 - Exposure to environmental and/or industrial chemical agents
 - Symptoms (if applicable) including right upper quadrant pain, hypersensitivity-type reactions, fatigue, nausea, vomiting and fever
 - Prior and/or concurrent use of alcohol, recreational drugs and special diets
 - Concomitant use of medications (including non-prescription medicines and herbal and dietary supplements), plants, and mushrooms
- Viral serologies
- CPK, haptoglobin, LDH, and peripheral blood smear
- Appropriate liver imaging if clinically indicated
- Appropriate blood sampling for pharmacokinetic analysis if this has not already been collected
- Hepatology consult (liver biopsy may be considered in consultation with an hepatologist)
- Follow the subject and the laboratory tests (ALT, AST, TBL, INR) until all laboratory abnormalities return to baseline or normal or considered stable by the investigator. The “close observation period” is to continue for a minimum of 4 weeks after discontinuation of all investigational product(s) and protocol-required therapies.

The potential DILI event and additional information such as medical history, concomitant medications and laboratory results must be captured in corresponding CRFs.

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Appendix B. eSerious Event Contingency Form

Completion Instructions - Electronic Adverse Event Contingency Report Form (For use for clinical trial studies using Electronic Data Capture [EDC])

NOTE: This form is to be used under restricted conditions outlined on page 1 below. If you must fax an event report to Amgen, you must also enter that event into the EDC system (eg, Rave) when it becomes available.

General Instructions

The protocol will provide instruction on what types of events to report for the study. This form is to be used **ONLY** to report events that must be captured in the Amgen safety database. *Indicates a mandatory field.

Types of Events to be reported on this form

- Serious Adverse Events (regardless of causal relationship to IP)

1. Site Information

Site Number* – Enter your assigned site number for this study

Investigator*, Country*, Reporter*, Phone No., and Fax No. – Enter information requested

2. Subject Information

Subject ID Number* – Enter the entire number assigned to the subject

Age at event onset, Sex, and Race – Enter the subject's demographic information

End of Study date – If the subject has already completed the study or terminated the study early, enter the End of Study date

If you are submitting follow-up information to a previous report, provide the serious adverse event term for the previous report as well as the start date for the initial event.

3. Serious Adverse Event

Provide the date the Investigator became aware of this Information

Serious Adverse Event Diagnosis or Syndrome* –

- If the diagnosis is known, it should be entered. Do not list all signs/symptoms if they are included in the diagnosis.
- If a diagnosis is not known, the relevant signs/symptoms should be entered.
- If the event is fatal, the cause of death should be entered and autopsy results should be submitted, when available.

Date Started* – Enter date the adverse event first started (not the date on which the event met serious criteria) rather than the date of diagnosis or hospitalization. **This is a mandatory field.**

Date Ended – Enter date the adverse event ended and not the date when the event no longer met serious criteria. If the event has not ended at the time of the initial report, a follow-up report should be completed when the end date is known. If the event is fatal, enter the date of death as the end date.

If event occurred before the first dose of Investigational Product (IP)/drug under study, add a check mark in the corresponding box.

Is event serious?* – Indicate Yes or No. **This is a mandatory field.**

Serious Criteria Code* – **This is a mandatory field for serious events.** Enter all reasons why the reported event has met serious criteria:

- Immediately life-threatening – Use only if the subject was at immediate risk of death from the event as it occurred. Emergency treatment is often required to sustain life in this situation.
- If the investigator decides an event should be reported in an expedited manner, but it does not meet other serious criteria, "Other Medically Important Serious Event" may be the appropriate serious criterion.

Relationship to IP – The Investigator must determine and enter the relationship of the event to the IP at the time the event is initially reported. **This is a mandatory field.**

Relationship to Amgen device* – The Investigator must determine and enter the relationship of the event to the Amgen device (e.g. prefilled syringe, auto-injector) at the time the event is initially reported. **If the study involves an Amgen device, this is a mandatory field. This question does not apply to non-Amgen devices used in the study (e.g. heating pads, infusion pumps)**

Outcome of Event* – Enter the code for the outcome of the event at the time the form is completed. **This is a mandatory field.**

- Resolved – End date is known
- Not resolved / Unknown – End date is unknown
- Fatal – Event led to death

If event is related to a study procedure, such as a biopsy, radiotherapy or withdrawal of a current drug treatment during a wash-out period, add a check mark to the corresponding box. This does not include relationship to IP or concomitant medication – only diagnostic tests or activities mandated by the protocol.

4. Hospitalization

If the subject was hospitalized, enter admission and discharge dates. Hospitalization is any in-patient hospital admission for medical reasons, including an overnight stay in a healthcare facility, regardless of duration. A pre-existing condition that did

**Completion Instructions - Electronic Adverse Event Contingency Report Form
(for use for Studies using Electronic Data Capture [EDC])**

Note, this form is to be used under restricted conditions outlined on page 1 of the form. If you must fax an event report to Amgen, you must also enter that event into the EDC system (eg, Rave) when it becomes available.

not worsen while on study which involved a hospitalization for an elective treatment, is not considered an adverse event. Protocol specified hospitalizations are exempt.

At the top of Page 2, provide your Site Number and the Subject ID Number in the designated section.

5. IP Administration including Lot # and Serial # when known / available.

Blinded or open-label – If applicable, indicate whether the investigational product is blinded or open-label

Initial Start Date – Enter date the product was first administered, regardless of dose.

Date of Dose Prior to or at the time of the Event – Enter date the product was last administered prior to, or at the time of, the onset of the event.

Dose, Route, and Frequency at or prior to the event – Enter the appropriate information for the dose, route and frequency at, or prior to, the onset of the event.

Action Taken with Product – Enter the status of the product administration.

6. Concomitant Medications

Indicate if there are any medications.

Medication Name, Start Date, Stop Date, Dose, Route, and Frequency – Enter information for any other medications the subject is taking. Include any study drugs not included in section 5 (Product Administration) such as chemotherapy, which may be considered co-suspect.

Co-suspect – Indicate if the medication is co-suspect in the event

Continuing – Indicate if the subject is still taking the medication

Event Treatment – Indicate if the medication was used to treat the event

7. Relevant Medical History

Enter medical history that is relevant to the reported event, not the event description. This may include pre-existing conditions that contributed to the event allergies and any relevant prior therapy, such as radiation. Include dates if available.

8. Relevant Laboratory Tests

Indicate if there are any relevant laboratory values.

For each test type, enter the test name, units, date the test was run and the results.

9. Other Relevant Tests

Indicate if there are any tests, including any diagnostics or procedures.

For each test type, enter the date, name, results and units (if applicable).

At the top of Page 3, provide your Site Number and the Subject ID Number in the designated section.

10. Case Description

Describe Event – Enter summary of the event. Provide narrative details of the events listed in section 3. Include any therapy administered, such as radiotherapy; (excluding medications, which will be captured in section 6). If necessary, provide additional pages to Amgen.

Complete the signature section at the bottom of page 3 and fax the form to Amgen. If the reporter is not the investigator, designee must be identified on the Delegation of Authority form.

AMGEN Study # 20150196 AMG 570	Electronic Serious Adverse Event Contingency Report Form For Restricted Use								
Reason for reporting this event via fax The Clinical Trial Database (eg. Rave): ☐ Is not available due to internet outage at my site ☐ Is not yet available for this study ☐ Has been closed for this study									
US: +888 814 8653									
1. SITE INFORMATION									
Site Number 	Investigator 	Country 							
Reporter 		Phone Number () 	Fax Number () 						
2. SUBJECT INFORMATION		Subject ID Number 		Age at event onset 	Sex ☐ F ☐ M	Race 	If applicable, provide End of Study date 		
If this is a follow-up to an event reported in the EDC system (eg. Rave), provide the adverse event term: _____ and start date: Day _____ Month _____ Year _____									
3. SERIOUS ADVERSE EVENT									
Provide the date the investigator became aware of this information: Day _____ Month _____ Year _____									
Serious Adverse Event <u>diagnosis</u> or syndrome If diagnosis is unknown, enter signs / symptoms and provide diagnosis, when known, in a follow-up report. List one event per line. If event is fatal, enter the cause of death. Entry of "death" is not acceptable, as this is an outcome.		Date Started 	Date Ended 	Check only if event occurred before first dose of IP ☐ Yes ☐ No	Is event serious? ☐ Yes ☐ No	Is event serious? (see codes below) ☐ Yes ☐ No	Relationship Is there a reasonable possibility that the event may have been caused by IP or an Amgen device used to administer the IP? 	Outcome of Event ☐ Resolved ☐ Not resolved ☐ Fatal ☐ Unknown	Check only if event is related to study procedure eg. biopsy ☐ Yes ☐ No
Serious Criteria: 01 Fatal 02 Immediately life-threatening		03 Required/prolonged hospitalization 04 Persistent or significant disability /incapacity		05 Congenital anomaly / birth defect 06 Other medically important serious event					
4. Was subject hospitalized or was a hospitalization prolonged due this event? ☐ No ☐ Yes If yes, please complete all of Section 4									
Date Admitted Day _____ Month _____ Year _____				Date Discharged Day _____ Month _____ Year _____					
5. Was IP/drug under study administered/taken prior to this event? ☐ No ☐ Yes If yes, please complete all of Section 5									
IP/Amgen Device:		Date of Initial Dose Day _____ Month _____ Year _____	Date of Dose Day _____ Month _____ Year _____	Dose 	Route 	Frequency 	Action Taken with Product 01 Still being Administered 02 Permanently discontinued 03 Withheld	Lot # and Serial # Lot # _____ Serial # _____ ☐ Unavailable / Unknown	
AMG 570		☒ blinded ☐ open label							

FORM-056006

Page 1 of 3

Version 7.0 Effective Date: 1 February 2016

AMGEN Study # 20150196 AMG 570	Electronic Serious Adverse Event Contingency Report Form For Restricted Use
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		Site Number			Subject ID Number											
6. CONCOMITANT MEDICATIONS (eg, chemotherapy) Any Medications? ☐ No ☐ Yes If yes, please complete:																
Medication Name(s)		Start Date			Stop Date			Co-suspect		Continuing		Dose	Route	Freq.	Treatment Med	
		Day	Month	Year	Day	Month	Year	No	Yes	No	Yes				No	Yes
7. RELEVANT MEDICAL HISTORY (include dates, allergies and any relevant prior therapy)																
8. RELEVANT LABORATORY VALUES (include baseline values) Any Relevant Laboratory values? ☐ No ☐ Yes If yes, please complete:																
Date	Test															
	Unit															
Day Month Year																
9. OTHER RELEVANT TESTS (diagnostics and procedures) Any Other Relevant tests? ☐ No ☐ Yes If yes, please complete:																
Date		Additional Tests						Results				Units				
Day Month Year																

Approved

[illegible]

Approved

Appendix C. Pregnancy and Lactation Notification Worksheets

AMGEN[®] Pregnancy Notification Worksheet

Fax Completed Form to the Country-respective Safety Fax Line

SELECT OR TYPE IN A FAX#

1. Case Administrative Information

Protocol/Study Number: 20150196

Study Design: ☒ Interventional ☐ Observational (If Observational: ☐ Prospective ☐ Retrospective)

2. Contact Information

Investigator Name _____ Site # _____
Phone (____) _____ Fax (____) _____ Email _____
Institution _____
Address _____

3. Subject Information

Subject ID # _____ Subject Gender: ☐ Female ☐ Male Subject DOB: mm ____ / dd ____ / yyyy ____

4. Amgen Product Exposure

Amgen Product	Dose at time of conception	Frequency	Route	Start Date
				mm ____ / dd ____ / yyyy ____

Was the Amgen product (or study drug) discontinued? ☐ Yes ☐ No

If yes, provide product (or study drug) stop date: mm ____ / dd ____ / yyyy ____

Did the subject withdraw from the study? ☐ Yes ☐ No

5. Pregnancy Information

Pregnant female's LMP mm ____ / dd ____ / yyyy ____ ☐ Unknown

Estimated date of delivery mm ____ / dd ____ / yyyy ____ ☐ Unknown ☐ N/A

If N/A, date of termination (actual or planned) mm ____ / dd ____ / yyyy ____

Has the pregnant female already delivered? ☐ Yes ☐ No ☐ Unknown ☐ N/A

If yes, provide date of delivery: mm ____ / dd ____ / yyyy ____

Was the infant healthy? ☐ Yes ☐ No ☐ Unknown ☐ N/A

If any Adverse Event was experienced by the infant, provide brief details: _____

Form Completed by:

Print Name: _____ Title: _____
Signature: _____ Date: _____

Approved

AMGEN Lactation Notification Worksheet

Fax Completed Form to the Country-respective Safety Fax Line

SELECT OR TYPE IN A FAX#

enter fax number

1. Case Administrative Information

Protocol/Study Number:

Study Design: ☐ Interventional ☐ Observational (If Observational: ☐ Prospective ☐ Retrospective)

2. Contact Information

Investigator Name

Site #

Phone ()

Fax ()

Email

Institution

Address

3. Subject Information

Subject ID #

Subject Date of Birth: mm / dd / yyyy

4. Amgen Product Exposure

Amgen Product	Dose at time of breast feeding	Frequency	Route	Start Date
				mm / dd / yyyy

Was the Amgen product (or study drug) discontinued? ☐ Yes ☐ No

If yes, provide product (or study drug) stop date: mm / dd / yyyy

Did the subject withdraw from the study? ☐ Yes ☐ No

5. Breast Feeding Information

Did the mother breastfeed or provide the infant with pumped breast milk while actively taking an Amgen product? ☐ Yes ☐ No

If No, provide stop date: mm / dd / yyyy

Infant date of birth: mm / dd / yyyy

Infant gender: ☐ Female ☐ Male

Is the infant healthy? ☐ Yes ☐ No ☐ Unknown ☐ N/A

If any Adverse Event was experienced by the mother or the infant, provide brief details:

Form Completed by:

Print Name:

Title:

Signature:

Date:

Appendix D. Patient Global Assessment of Disease Activity and Disability Index of the Health Assessment Question (HAQ-DI)

AMGEN AMG 570 20150196	Site No.				Subject ID No.							
					1	9	6					

PATIENT GLOBAL ASSESSMENT OF DISEASE ACTIVITY

Considering all of the possible symptoms of rheumatoid arthritis (RA), please draw a vertical bar (|) through the line below to indicate how active your RA has been this past week.

0 |-----| 100
 No RA Activity At All Worst RA Activity Imaginable

<i>For Study Site Staff Use Only</i>	SCORE (in mm)	

For Study Site Staff Use Only

DATE		
Day	Month	Year

AMGEN AMG 570 20150196	Site No.				Subject ID No.							
					1	9	6					

HEALTH ASSESSMENT QUESTIONNAIRE

In this section we are interested in learning how your illness affects your ability to function in daily life. Please feel free to add any comments on the back of this page.

Please check the response which best describes your usual abilities OVER THE PAST WEEK:

	<u>Without ANY Difficulty</u>	<u>With SOME Difficulty</u>	<u>With MUCH Difficulty</u>	<u>UNABLE To Do</u>
DRESSING & GROOMING				
Are you able to:				
- Dress yourself, including tying shoelaces and doing buttons?	_____	_____	_____	_____
- Shampoo your hair?	_____	_____	_____	_____
ARISING				
Are you able to:				
- Stand up from a straight chair?	_____	_____	_____	_____
- Get in and out of bed?	_____	_____	_____	_____
EATING				
Are you able to:				
- Cut your meat?	_____	_____	_____	_____
- Lift a full cup or glass to your mouth?	_____	_____	_____	_____
- Open a new milk carton?	_____	_____	_____	_____
WALKING				
Are you able to:				
- Walk outdoors on flat ground?	_____	_____	_____	_____
- Climb up five steps?	_____	_____	_____	_____

Please check any AIDS OR DEVICES that you usually use for any of these activities:

_____ Cane	_____ Devices used for dressing (button hook, zipper pull, long-handled shoe horn, etc.)
_____ Walker	_____ Built up or special utensils
_____ Crutches	_____ Special or built up chair
_____ Wheelchair	_____ Other (Specify: _____)

Please check any categories for which you usually need HELP FROM ANOTHER PERSON:

_____ Dressing and Grooming	_____ Eating
_____ Arising	_____ Walking

Approved

 AMG 570 20150196	Site No.				Subject ID No.							
					1	9	6					

Please check the response which best describes your usual abilities OVER THE PAST WEEK:

	Without ANY Difficulty	With SOME Difficulty	With MUCH Difficulty	UNABLE To Do
HYGIENE				
Are you able to:				
- Wash and dry your body?	_____	_____	_____	_____
- Take a tub bath?	_____	_____	_____	_____
- Get on and off the toilet?	_____	_____	_____	_____
REACH				
Are you able to:				
- Reach and get down a 5 pound object (such as a bag of sugar) from just above your head?	_____	_____	_____	_____
- Bend down to pick up clothing from the floor?	_____	_____	_____	_____
GRIP				
Are you able to:				
- Open car doors?	_____	_____	_____	_____
- Open jars which have been previously opened?	_____	_____	_____	_____
- Turn faucets on and off?	_____	_____	_____	_____
ACTIVITIES				
Are you able to:				
- Run errands and shop?	_____	_____	_____	_____
- Get in and out of a car?	_____	_____	_____	_____
- Do chores such as vacuuming or yardwork?	_____	_____	_____	_____

Please check any AIDS OR DEVICES that you usually use for any of these activities:

☐ Raised toilet seat	☐ Bathtub bar
☐ Bathtub seat	☐ Long-handled appliances for reach
☐ Jar opener (for jars previously opened)	☐ Long-handled appliances in bathroom
	☐ Other (Specify: _____)

Please check any categories for which you usually need HELP FROM ANOTHER PERSON:

☐ Hygiene	☐ Gripping and opening things
☐ Reach	☐ Errands and chores

We are also interested in learning whether or not you are affected by pain because of your illness.

How much pain have you had because of your illness IN THE PAST WEEK:

PLACE A VERTICAL (I) MARK ON THE LINE TO INDICATE THE SEVERITY OF THE PAIN.

NO PAIN 0	_____	SEVERE PAIN 100
-----------------	-------	-----------------------

Amendment 3

Protocol Title: A Randomized, Double Blind Placebo Controlled Study to Evaluate the Safety, Tolerability, and Pharmacokinetics of Multiple Ascending Subcutaneous Doses of AMG 570 in Subjects With Rheumatoid Arthritis

Amgen Protocol Number (AMG 570) 20150196

IND number BB-IND 127060

EudraCT number 2017-000337-31

Amendment Date: 08 June 2018

Rationale:

The protocol has been amended to include an additional cohort – Cohort 4. Enrollees in Cohort 4 will receive multiple subcutaneous (SC) doses of 420 mg AMG 570 IP. Subjects will receive a SC dose of IP on days 1, 15, 29, 43, 57, and 71 (every 2 weeks for 6 doses). The addition of this cohort will extend enrollment to an additional 8 subjects for a total of 32 subjects and will provide additional safety, pharmacokinetics, and pharmacodynamics data that will facilitate the understanding of the exposure-response relationship to inform future studies with AMG 570. Available data from the single ascending dose (SAD) study with AMG 570 (20140322) have been used to support updates to dose selection, study design, safety and biostatistical information throughout the protocol. Where available, observed data have replaced predicted values for increased accuracy.

Additional changes have been made for increased clarity throughout the protocol where applicable.

Administrative, typographical and formatting changes were made throughout the protocol where applicable.

Amendment 2

Protocol Title: A Randomized, Double Blind Placebo Controlled Study to Evaluate the Safety, Tolerability, and Pharmacokinetics of Multiple Ascending Subcutaneous Doses of AMG 570 in Subjects With Rheumatoid Arthritis

Amgen Protocol Number (AMG 570) 20150196

IND number BB-IND 127060

EudraCT number 2017-000337-31

Amendment Date: 13 November 2017

Rationale:

The protocol has been amended to make revisions to laboratory assessments, eligibility criteria, and biomarker tests.

Laboratory assessments:

Changes have been made to hematology labs in light of findings outlined in the safety report submitted on 15-Nov-2017.

1. Reticulocyte counts will now be assessed along with CBC with differential.
2. Hematology labs (CBC with differential and reticulocyte count) will be examined at all outpatient visits beginning on Day 8. After day 8, hematology monitoring will occur at least every 2 weeks during treatment period and hematology parameters will continue to be monitored until the end of study at day 239.
3. RBCs are named parameters of interest in the CBC with differential.

Eligibility criteria:

Exclusion Criterion 223 has been revised to better account for population of interest and ensure medical monitor input prior to enrollment. The lower limit of normal for the assay being employed in study 20150196 for absolute B cell counts in the peripheral blood is 107 B cells per μL . Though absolute B cell counts in patients with rheumatoid arthritis (RA) are on average in the normal, healthy range, the counts tend to be lower than healthy controls. In a study of 208 RA patients prior to receiving rituximab, Sellam, et. al. (Arthritis & Rheumatism 63: 3692), report that RA patients averaged 188.6 B cells per μL compared to 257.3 B cells per μL in healthy controls. Thus, B cell numbers in patients with RA are ~70% of healthy controls. Despite tending to have lower B cell numbers, patients with RA mount protective antibody responses to

vaccination against pneumococcus or influenza virus at rates of 70-98% (BMC Musculoskelet Disord 17: 231 and J Rheumatology 34: 272). In addition, RA patients treated with rituximab who have evidence of normal B cell function assessed by IgG levels benefit from complete peripheral B cell depletion with an acceptable safety profile (Arthritis & Rheumatism 54: 2793). Thus, changes to the eligibility criteria related to B cell numbers would be considered safe in the context of up-to-date vaccine status (inclusion criterion 108), normal antibody levels (exclusion criterion 224), and a lack of history of recurrent/chronic infections (exclusion criteria 206).

Biomarker tests:

Biomarker testing has been changed from plasma cell by flow cytometry to Tfh by flow cytometry. There has been no change to the sampling schedule or to the amount of blood and type of tube to be drawn.

Additional changes have been made for increased clarity throughout the protocol where applicable.

Administrative, typographical and formatting changes were made throughout the protocol where applicable.

Approved

Amendment 1

Protocol Title: A Randomized, Double Blind Placebo Controlled Study to Evaluate the Safety, Tolerability, and Pharmacokinetics of Multiple Ascending Subcutaneous Doses of AMG 570 in Subjects With Rheumatoid Arthritis

Amgen Protocol Number AMG 570 20150196

Amendment Date: 16 May 2017

Rationale:

The protocol has been amended to make corrections to required laboratory assessments prior to first dose.

- Laboratory assessments have been added to applicable protocol sections:
 - Pre-dose C-Reactive Protein (CRP) is a required assessment which was omitted in error.
 - Pre-dose Erythrocyte Sedimentation Rate (ESR) is a required assessment which was omitted in error.
- Laboratory assessments deemed not technically feasible have been removed from applicable protocol sections:
 - Pre-dose Plasma Cell by flow cytometry

Additional changes have been made for increased clarity throughout the protocol where applicable.

Administrative, typographical and formatting changes were made throughout the protocol where applicable.

Approved